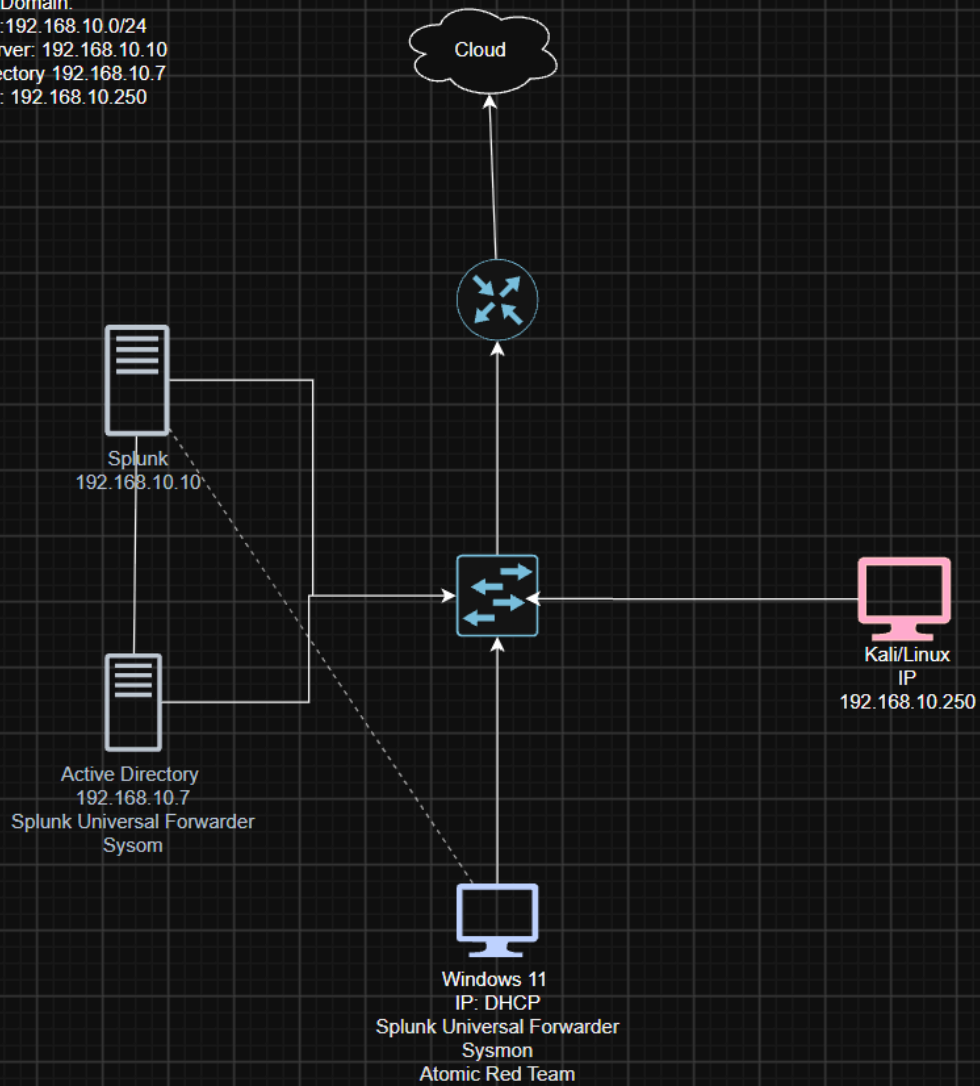


I started by setting up a diagram of the systems I will be using to complete the lab (made on draw.io)

- Splunk and active directory server being connected to the switch
- Windows 11 connected to Splunk sever and router
- Kali Linux attacking server isolated and only connected to switch
- Switch connected to router
- router connected to Cloud
- Atomic Red Team runs small MITRE ATT&CK-mapped attack simulations from the Kali box to generate realistic malicious activity for Splunk to detect.
- The Splunk Universal Forwarder is installed on the Windows 11 and AD hosts to collect logs (including Sysmon) and forward them to the Splunk server for indexing.

Domain:
Network: 192.168.10.0/24
Splunk Server: 192.168.10.10
Active Directory: 192.168.10.7
Attacker: 192.168.10.250



The

SETUP for Windows 11 (not used)

Windows 11 Installation Assistant

This is the best option for installing Windows 11 on the device you're using.

Note: Windows 11 Installation Assistant doesn't run on Arm-based PCs.

> Before you begin using Installation Assistant

[Download Now](#)

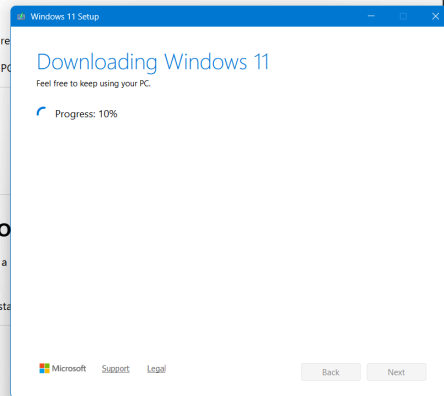
Create Windows 11 Installation Media

If you want to perform a reinstall or clean install of Windows 11 on a bootable USB or DVD.

Note: Windows 11 media creation tool cannot be used to create installation media for Arm-based PCs.

> Before you begin using the media creation tool

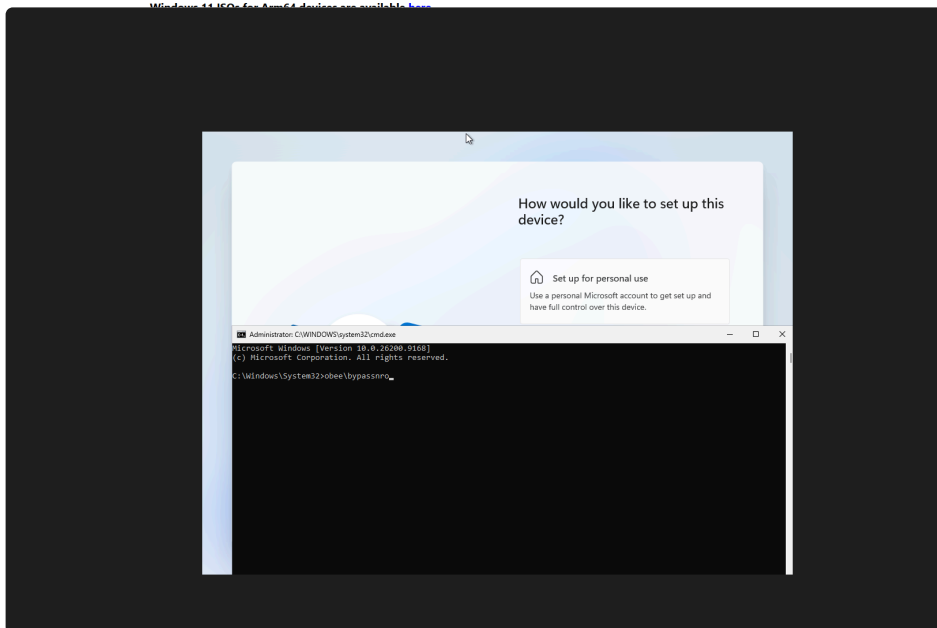
[Download Now](#)



Download Windows 11 Disk Image (ISO) for x64 devices

This option is for users that want to create a bootable installation media (USB flash drive, DVD) or create a virtual machine (ISO file) to install Windows 11. This download is a multi-edition ISO which uses your product key to unlock the correct edition.

[Windows 11 ISOs for Arm64 devices are available here.](#)



While in setup you can skip the account set up option by using shift F10 and running the oobe\bypassnro command

The command makes the VM reboot itself and drop it will put us back at the network selection screen. this time with an option to select "I don't have internet" and then "Continue with limited setup"

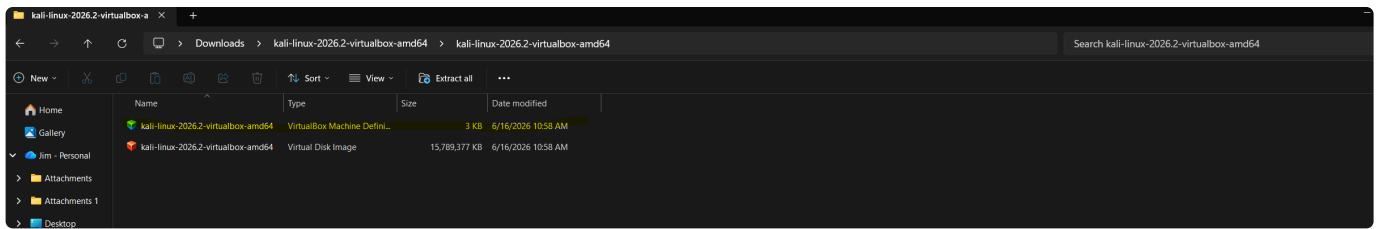
From there it lets you create a local account (username/password only, no Microsoft account, no internet needed)

However for my lab i decided to use windows 10 due to convience.

Kali Linux setup

<https://www.kali.org/get-kali/#kali-virtual-machines>

Kali Linux downloading process was straight forward, once on the website i downloaded the virtual box option



I double clicked the top option that was inside my \Downloads\kali-linux-2026.2-virtualbox-amd64.7z\kali-linux-2026.2-virtualbox-amd64

it automatically took me into VirtualBox where I was easily able to configure settings for storage and ram

Windows 10

Continue where you left off: Chrome can restore your tabs every time you restart. To turn this on, go to [Settings](#).

Stay Secure with Essential Windows Updates: After October 14, 2025, Windows 10 will no longer receive free software updates, technical support, or security fixes. This means your PC will be more vulnerable to security threats and malware. **Consider Upgrading to Windows 11:** Move to the security, speed, and innovation that Windows 11 PCs provide, available at every price point. Upgrading to Windows 11 ensures you continue to receive the latest security updates, features, and technical support, keeping your PC safe and efficient. For more information on preparing for Windows 10 end of support, see our [Windows blog post](#).

Windows 10 2022 Update | Version 22H2

The Update Assistant can help you update to the latest version of Windows 10. To get started, click **Update now**.

[Update now](#)

Create Windows 10 installation media

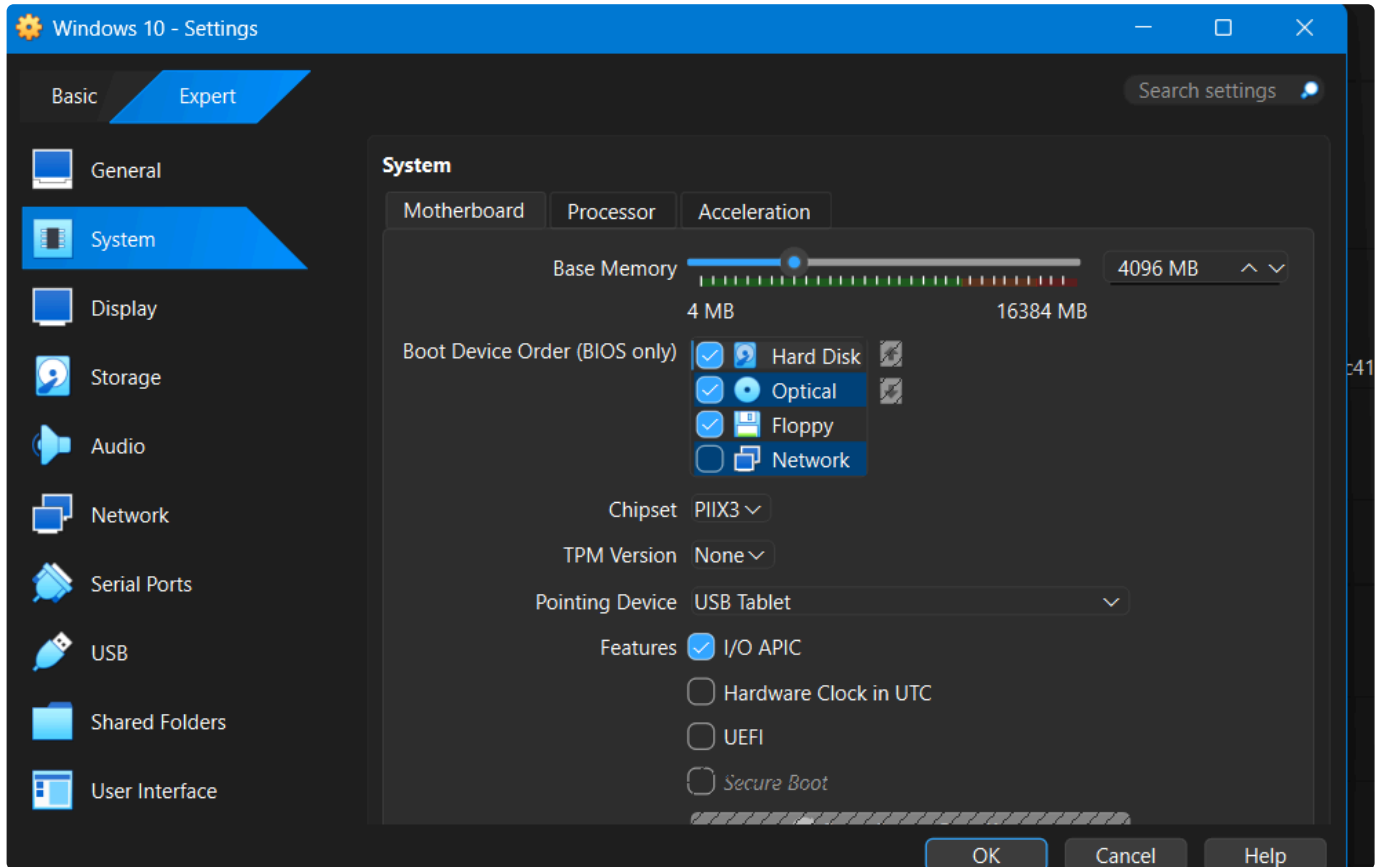
To get started, you will first need to have a license to install Windows 10. You can then download and run the media creation tool. For more information on how to use the tool, see the instructions below.

[Download Now](#)



Downloaded the "create installation media" and then use it to create an iso file.

open virtual box and uploaded the iso image into the add tool, then I gave it 4096mb with 2 CPUs



start the vm, install the windows 10 pro with and ensure that you run the custom installation over the upgrade.



Windows Setup

Select the operating system you want to install

Operating system	Architecture	Date modified
Windows 10 Home	x64	5/5/2023
Windows 10 Home N	x64	5/5/2023
Windows 10 Home Single Language	x64	5/5/2023
Windows 10 Education	x64	5/5/2023
Windows 10 Education N	x64	5/5/2023
Windows 10 Pro	x64	5/5/2023
Windows 10 Pro N	x64	5/5/2023

Description:
Windows 10 Pro

Next



Windows Setup

Which type of installation do you want?

Upgrade: Install Windows and keep files, settings, and applications

The files, settings, and applications are moved to Windows with this option. This option is only available when a supported version of Windows is already running on the computer.

Custom: Install Windows only (advanced)

The files, settings, and applications aren't moved to Windows with this option. If you want to make changes to partitions and drives, start the computer using the installation disc. We recommend backing up your files before you continue.

Windows Sever 2022

<https://www.microsoft.com/en-us/evalcenter/download-windows-server-2022>

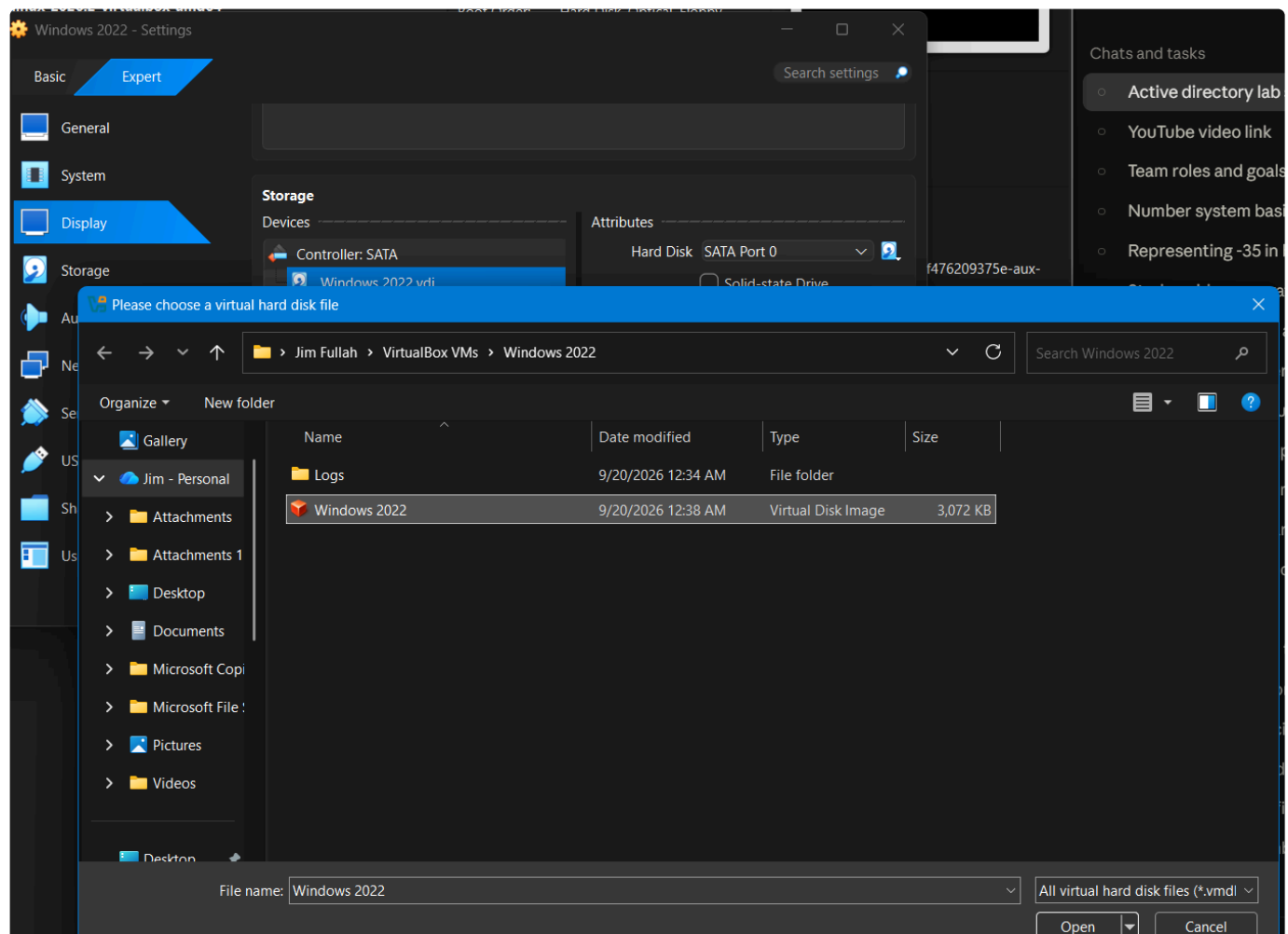
downloaded the ISO on the Microsoft website

repeated the VirtualBox Setup with default settings

Microsoft | Evaluation Center Windows Server SQL Server System Center Microsoft Security Additional products

Please select your Windows Server 2022 download

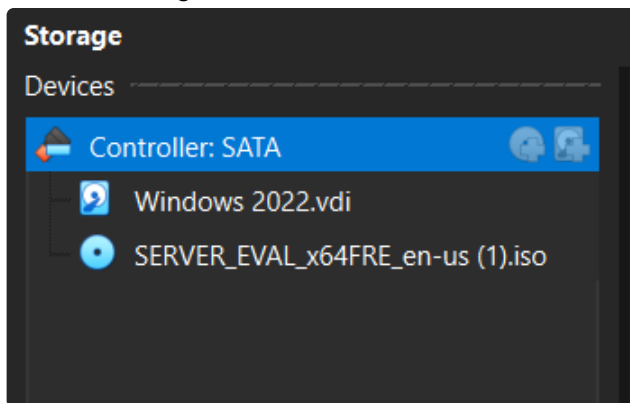
English (United States)	ISO downloads 64-bit edition >	VHD download 64-bit edition >	Try on Azure Learn more >	Create a VM in Azure Learn more >
Chinese (Simplified)	ISO downloads 64-bit edition >			
French	ISO downloads 64-bit edition >			
German	ISO downloads 64-bit edition >			
Italian				



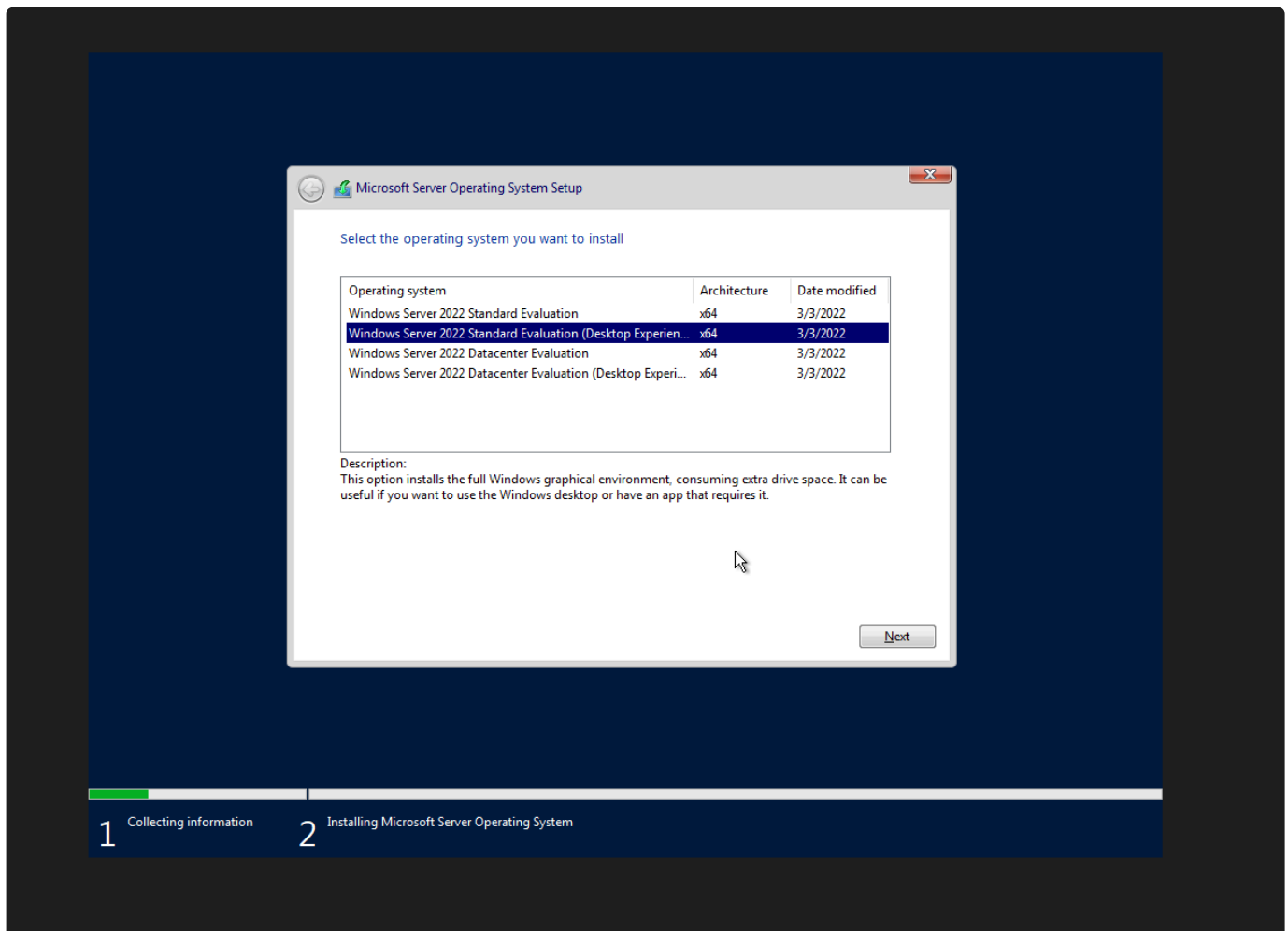
I went to Windows Server 2022 VM > Settings > Storage > click the Optical Drive icon (the small disc icon

at the top, next to "Controller: SATA") to add a new optical drive
clicked choose a disk file
Browse to your actual downloaded ISO
Click OK to save the VM settings

Before moving on assure that both Server Eval and the windows 2022 vdi are both attached



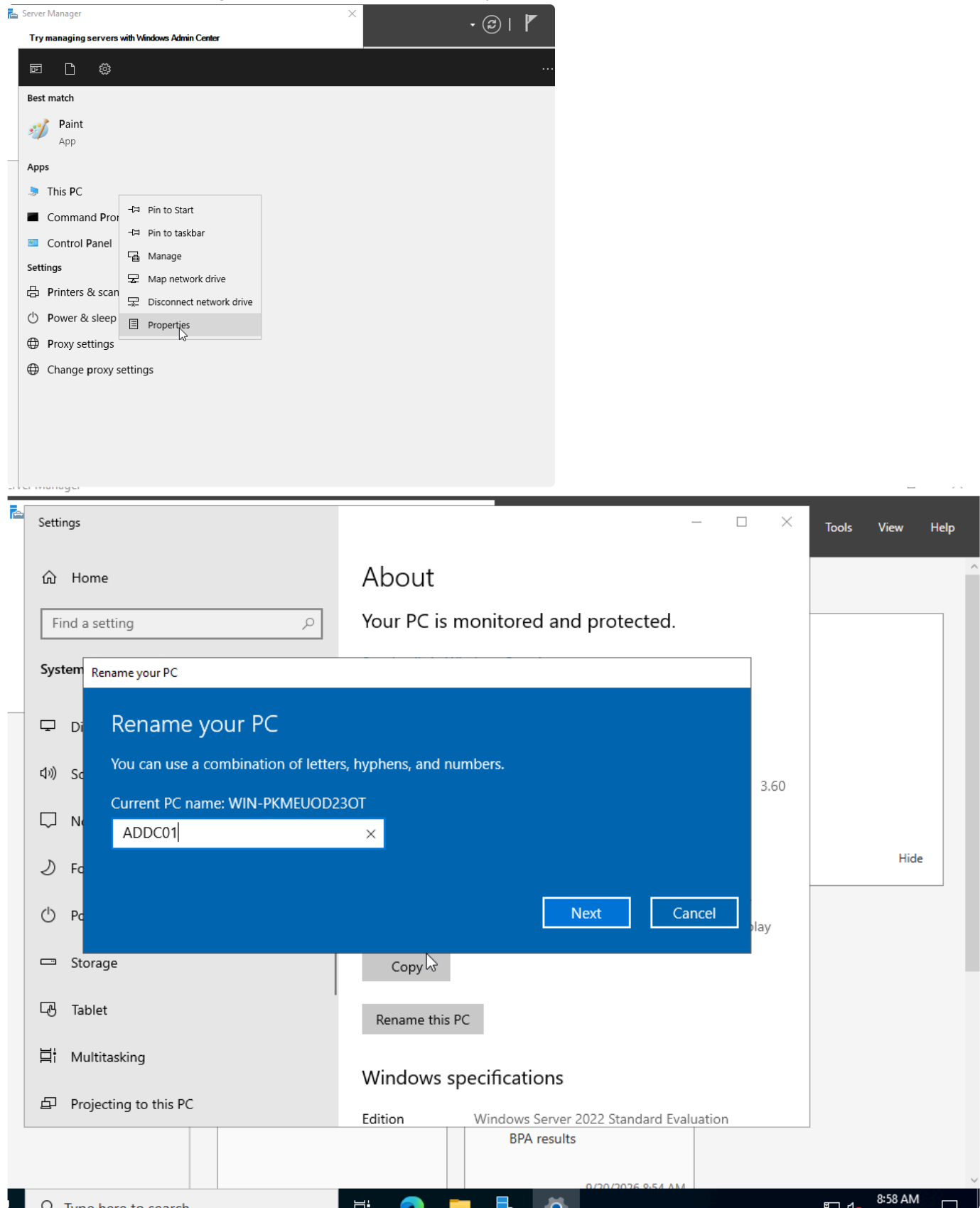
Proceed with standard evaluation desktop experience settings



Agree to terms of service and select the custom installation. Create an account and as a test log in on open server manger. We will be using server 2022 to install our active directory.

Lets change the servers name to match the diagram and to prevent confusion down the line.

Search up this pc then go to properties then rename your pc to "ADDC01" and restart



Splunk Server (ubuntu)

<https://ubuntu.com/download/server>

DownloadsDesktopServerWSLCoreCloud

BitTorrentOther images and mirrorsPast releasesNetwork installer

Past releases and other flavors

Looking for an older release of Ubuntu? Whether you need a POWER, IBMz (s390x), ARM, an obsolete release or a previous LTS point release with its original stack, you can find them in past releases.

Ubuntu 24.04 LTS Noble Numbat

Ubuntu 22.04 LTS Jammy Jellyfish

Ubuntu 20.04 LTS Focal Fossa

Standard support has ended for 20.04 LTS, now secured with [Ubuntu Pro](#)

Ubuntu 18.04 LTS Bionic Beaver

Standard support has ended for 18.04 LTS ended, now secured with [Ubuntu Pro](#)

Canonical UbuntuProductsUse casesSupportCommunityGet UbuntuCareersAll CanonicalSign in

DownloadsDesktopServerWSLCoreCloud

Thank you for downloading Ubuntu Server 26.04.1 LTS

Your download should start in the background. If it doesn't, [download now](#). You can [verify your download](#), or get [help on installing](#).

Ubuntu CLI cheatsheet

Learn how to use the command line efficiently and get started with DevOps — from basic file management to LXD virtualization and Ubuntu Pro.

Download the CLI cheat sheet

Ubuntu

System

System information

System monitoring and management

Networking commands

Service management

Core jobs and scheduling

Files

File management

Directory navigation

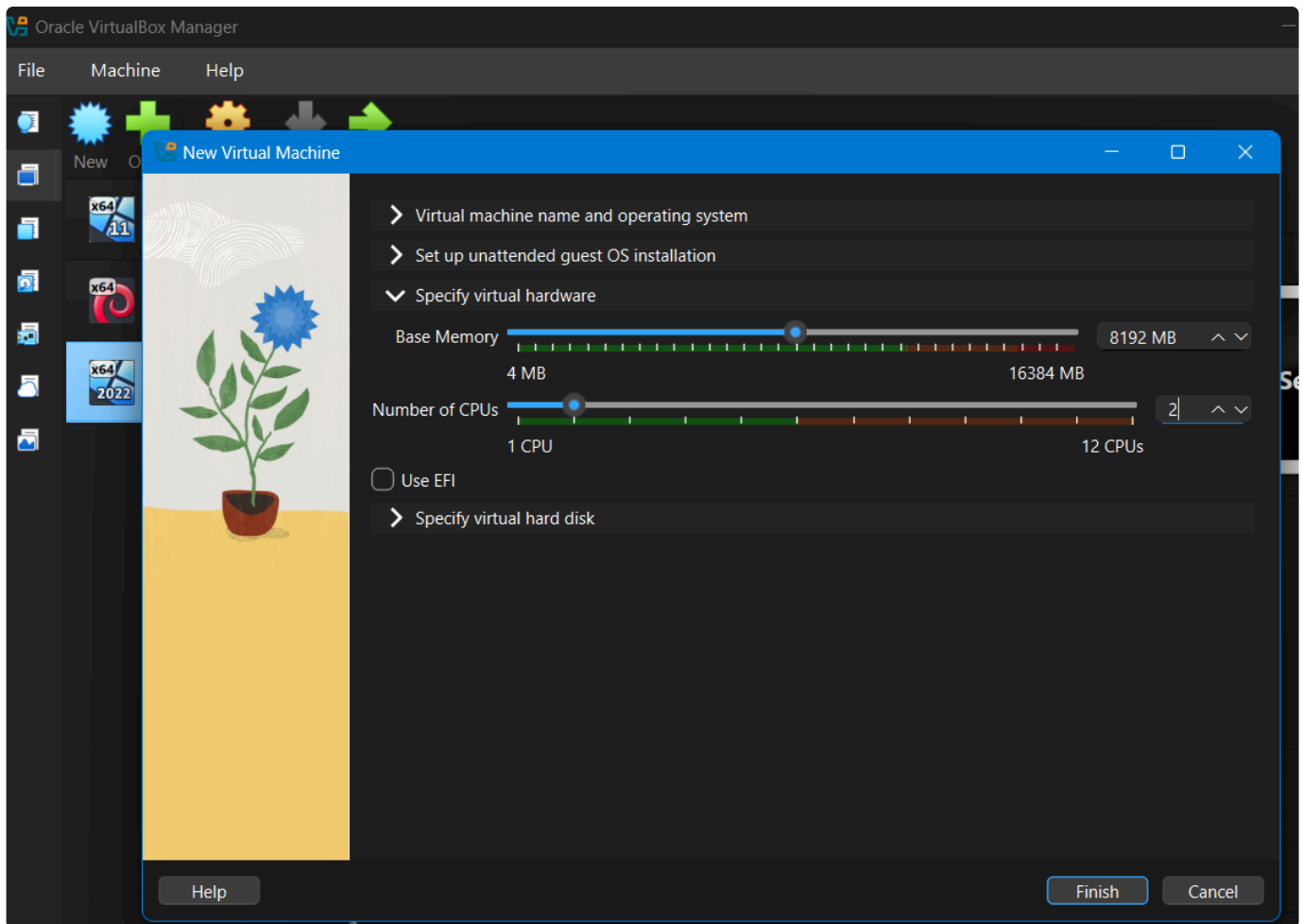
File permissions and ownership

Searching and finding

Archiving and compression

Text editing and processing

Setting up the splunk server with Additional space and ram since this will be capturing logs



In setup it will ask you to create a user and password, complete that then launch the vm.

```
vboxuser@Splunk0:~$ sudo apt-get update && sudo apt-get upgrade -y
[sudo: authenticate] Password:
Hit:1 http://us.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30 linux-headers-7.0.0-30-generic linux-image-unsigned-7.0.0-30-generic linux-main-modules-zfs-7.0.0-30-generic
  linux-modules-7.0.0-30-generic linux-tools-7.0.0-30 linux-tools-7.0.0-30-generic
Use 'sudo apt autoremove' to remove them.
The following upgrades have been deferred due to phasing:
  libaudit-common libaudit1 libflashrom1
The following packages will be upgraded:
  python3-apt-common python3-apt
2 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
Need to get 182 kB of archives.
After this operation, 4,096 B disk space will be freed.
Get:1 http://us.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-apt-common all 3.1.0ubuntu1.1 [17.0 kB]
Get:2 http://us.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-apt amd64 3.1.0ubuntu1.1 [165 kB]
Fetched 182 kB in 0s (910 kB/s)
(Reading database... 136592 files and directories currently installed.)
Preparing to unpack ./python3-apt-common_3.1.0ubuntu1.1_all.deb...
Unpacking python3-apt-common (3.1.0ubuntu1.1) over (3.1.0ubuntu1)...
Preparing to unpack ./python3-apt_3.1.0ubuntu1.1_amd64.deb...
Unpacking python3-apt (3.1.0ubuntu1.1) over (3.1.0ubuntu1)...
Setting up python3-apt-common (3.1.0ubuntu1.1)...
Setting up python3-apt (3.1.0ubuntu1.1)...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
  systemctl restart unattended-upgrades.service

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
vboxuser@Splunk0:~$
```

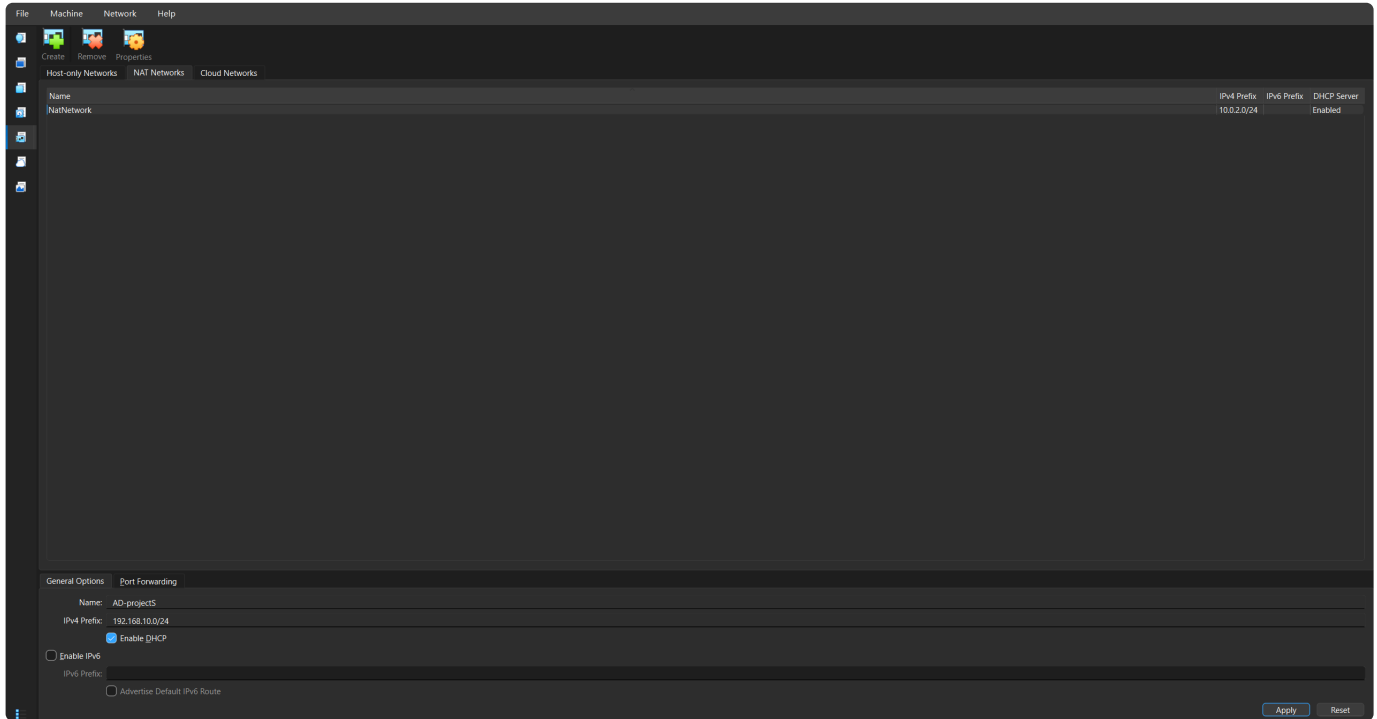
once installed I made sure the sever was up the date by using the apt-upgrade/update command

INSTALLING AND CONFIGURING SPLUNK & SYSMON

In this section I install Splunk Enterprise on the Ubuntu server, then set up the Universal Forwarder and Sysmon on the Windows 10 client. The goal is to get detailed endpoint activity flowing from the client into Splunk so it can be searched and analyzed later.

to ensure our VMS can be on the same network and still have internet access we will setup NAT network.

(NAT network is a process used by a router or firewall to change private IP addresses into a single public IP address so multiple devices can share one internet connection <https://www.cloudns.net/blog/nat-network-address-translation-how-it-works-and-why-it-matters/>)



(note the ipv4 prefix will be set up to match diagram)

briefly set all VMs network attachments to NAT network

First we will configure our Splunk server

```
vboxuser@Splunk0:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:c3:ef:0e brd ff:ff:ff:ff:ff:ff
    altname enx080027c3ef0e
    inet 192.168.10.3/24 metric 100 brd 192.168.10.255 scope global dynamic enp0s3
        valid_lft 585sec preferred_lft 585sec
    inet6 fe80::a00:27ff:fec3:ef0e/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
vboxuser@Splunk0:~$ ping google.com
PING google.com (172.253.122.113) 56(84) bytes of data:
64 bytes from bh-in-f113.1e100.net (172.253.122.113): icmp_seq=1 ttl=64 time=8.11 ms
64 bytes from bh-in-f113.1e100.net (172.253.122.113): icmp_seq=2 ttl=64 time=8.96 ms
64 bytes from bh-in-f113.1e100.net (172.253.122.113): icmp_seq=3 ttl=64 time=12.5 ms
64 bytes from bh-in-f113.1e100.net (172.253.122.113): icmp_seq=4 ttl=64 time=5.56 ms
^C
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 5969ms
rtt min/avg/max/mdev = 5.557/8.790/12.534/2.497 ms
vboxuser@Splunk0:~$
```

The server has been confirmed to reach websites, however we want our server to match the IP address in the diagram. To do so will give it a static IP address

Run "sudo nano /etc/netplan/00-installer-config.yaml"

```
vboxuser@Splunk0:~$ sudo nano /etc/netplan/00-installer-config.yaml _
```

Once the command is ran we want to

-set both DHCP false

(This turns off automatic IP assignment for both IPv4 and IPv6. It tells the system I will assign the address manually instead of letting the network hand one out.)

-add nameserverline

(This is the static IP I chose for the server, 192.168.10.10/24. The /24 sets the subnet mask, which tells the system how many devices can exist on this network segment.)

-add addresses line

(This tells the server which DNS server to use when it needs to resolve domain names. I pointed it at 8.8.8.8, which is Google's public DNS, so it can still reach the internet for updates and lookups.)

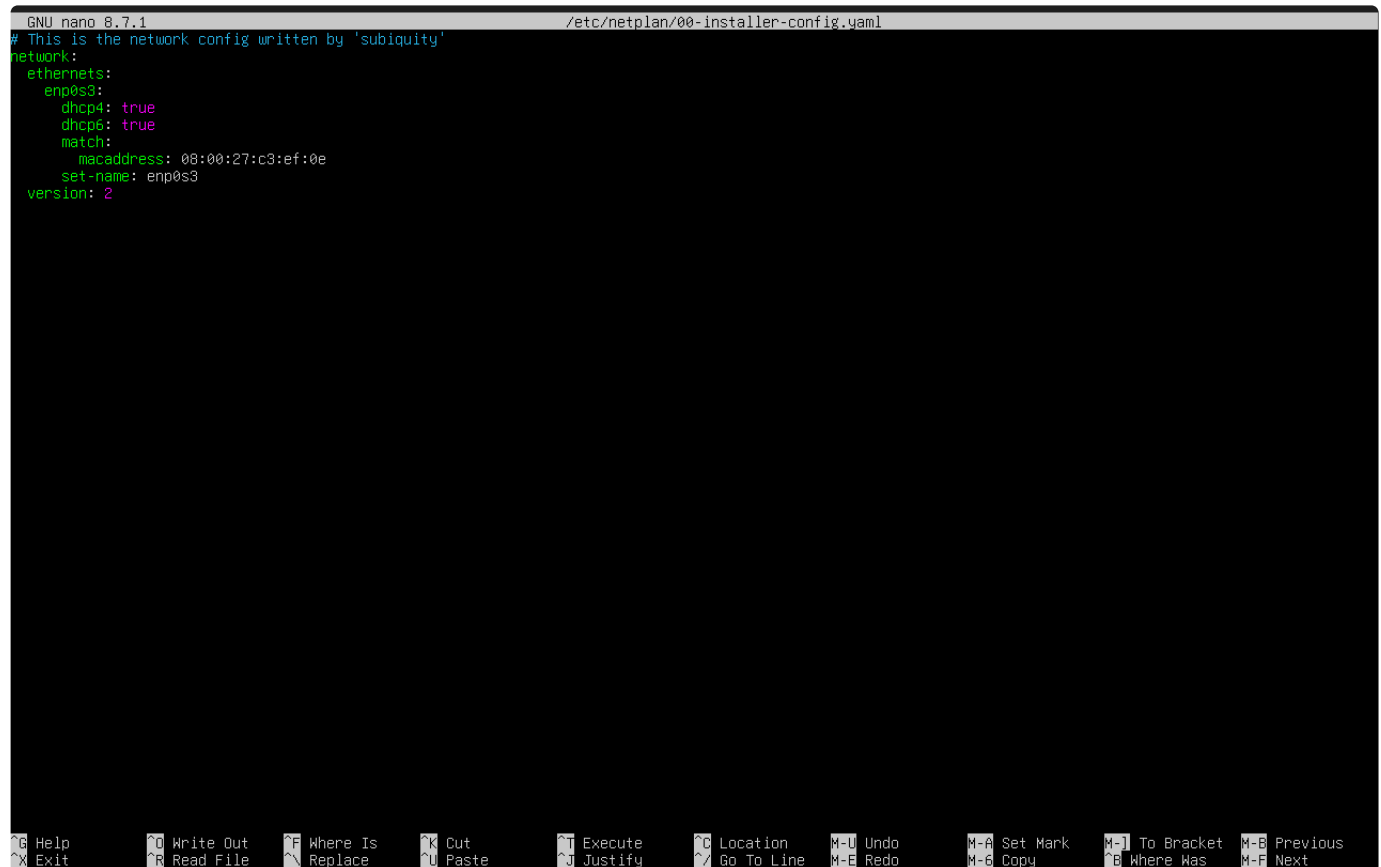
-add route sections with default route

(This defines where traffic goes when it is not meant for a device on the local network. The "to: default" line means this rule applies to all outgoing traffic that has no more specific route.)

-Added the via line (gateway)

This is the actual gateway address, 192.168.10.1. It is the device (usually the router or virtual network gateway) that forwards traffic out to other networks or the internet

```
GNU nano 8.7.1 /etc/netplan/00-installer-config.yaml
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: true
      dhcp6: true
      match:
        macaddress: 08:00:27:c3:ef:0e
      set-name: enp0s3
  version: 2
```



```
GNU nano 8.7.1 /etc/netplan/00-installer-config.yaml
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: no
      dhcp6: no
      addresses: [192.168.10.10/24]
      nameservers:
        addresses: [8.8.8.8]
      routes:
        - to: default
          via: 192.168.10.1
  version: 2
```

Run "sudo netplan apply" to apply changes

Confirm the change is in effect and the server can reach websites by running "ip a" and "ping google.com".

```
vboxuser@Splunk0:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:c3:ef:0e brd ff:ff:ff:ff:ff:ff
    altname enx080027c3ef0e
    inet 192.168.10.10/24 brd 192.168.10.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fec3:ef0e/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
vboxuser@Splunk0:~$ ping google.com
PING google.com (64.233.180.100) 56(84) bytes of data.
64 bytes from on-in-f100.1e100.net (64.233.180.100): icmp_seq=1 ttl=64 time=9.99 ms
64 bytes from on-in-f100.1e100.net (64.233.180.100): icmp_seq=2 ttl=64 time=9.63 ms
64 bytes from on-in-f100.1e100.net (64.233.180.100): icmp_seq=3 ttl=64 time=8.80 ms
64 bytes from on-in-f100.1e100.net (64.233.180.100): icmp_seq=4 ttl=64 time=9.19 ms
^C
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3026ms
rtt min/avg/max/mdev = 8.801/9.401/9.988/0.447 ms
vboxuser@Splunk0:~$ _
```

Downloading and installing Splunk

Head to the official Splunk website to create an account and download SplunkEnterprise using their free

Continue where you left off: Chrome can restore your tabs every time you restart. To turn this on, go to [Settings](#).

Splunk
A Cisco Company

PlatformSecurityObservabilityIndustriesResourcesPricing

PartnersSupport

Log InSign Up

Q

Trials & Downloads

Create your Splunk account

Operational Intelligence gives you a real-time understanding of what's happening across your IT systems and technology infrastructure so you can make informed decisions. It is enabled by the Splunk platform, the foundation for all of Splunk's products, premium solutions, apps and add-ons.

Free Trials and Downloads
Search, analyze and visualize the massive streams of machine data generated by your IT systems and technology infrastructure—physical, virtual and in the cloud.

Splunkbase Apps and Add-Ons
Apps from Splunk, our partners and our community enhance and extend the power of the Splunk platform.

Help When You Need It
Get answers to your questions from Splunk know-it-alls. We have tutorials, use cases, references,

Create Your Splunk Account

Enter your account details to continue to verification.

Business email

Continue

Splunk Enterprise 10.4.3

Index 500 MB/Day. Sign up and download now. After 60 days you can convert to a perpetual free license or purchase a Splunk Enterprise license to continue using the expanded functionality designed for enterprise-scale deployments.

Choose Your Installation Package

 Windows

 Linux

 Mac OS

64-bit	4.x+, 5.x+, 6.x+ kernel Linux distributions	.rpm	1531.55 MB	Download Now	Copy wget link	More ▾
		.tgz	1606.7 MB	Download Now	Copy wget link	More ▾
		.deb	1262.71 MB	Download Now	Copy wget link	More ▾

[Release Notes](#) | [System Requirements](#) | [Previous Releases](#) | [All Other Downloads](#)

I chose Splunk Enterprise's on premises option instead of a cloud instance because this lab is meant to simulate a self hosted environment, similar to what a small company might run in house rather than paying for a hosted SIEM.

Run back to the splunk server(ubuntu). On the command line run "sudo apt-get install virtualbox-guest-additions-iso"

```
vboxuser@Splunk0:~$ sudo apt-get install virtualbox-guest-additions-iso
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following NEW packages will be installed:
  virtualbox-guest-additions-iso
0 upgraded, 1 newly installed, 0 to remove and 3 not upgraded.
Need to get 39.0 MB of archives.
After this operation, 53.4 MB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu/resolute/multiverse/amd64 virtualbox-guest-additions-iso all 7.2.6-1 [39.0 MB]
Fetched 39.0 MB in 2s (24.9 MB/s)
Selecting previously unselected package virtualbox-guest-additions-iso.
(Reading database... 104264 files and directories currently installed.)
Preparing to unpack .../virtualbox-guest-additions-iso_7.2.6-1_all.deb...
Unpacking virtualbox-guest-additions-iso (7.2.6-1)...
Setting up virtualbox-guest-additions-iso (7.2.6-1)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

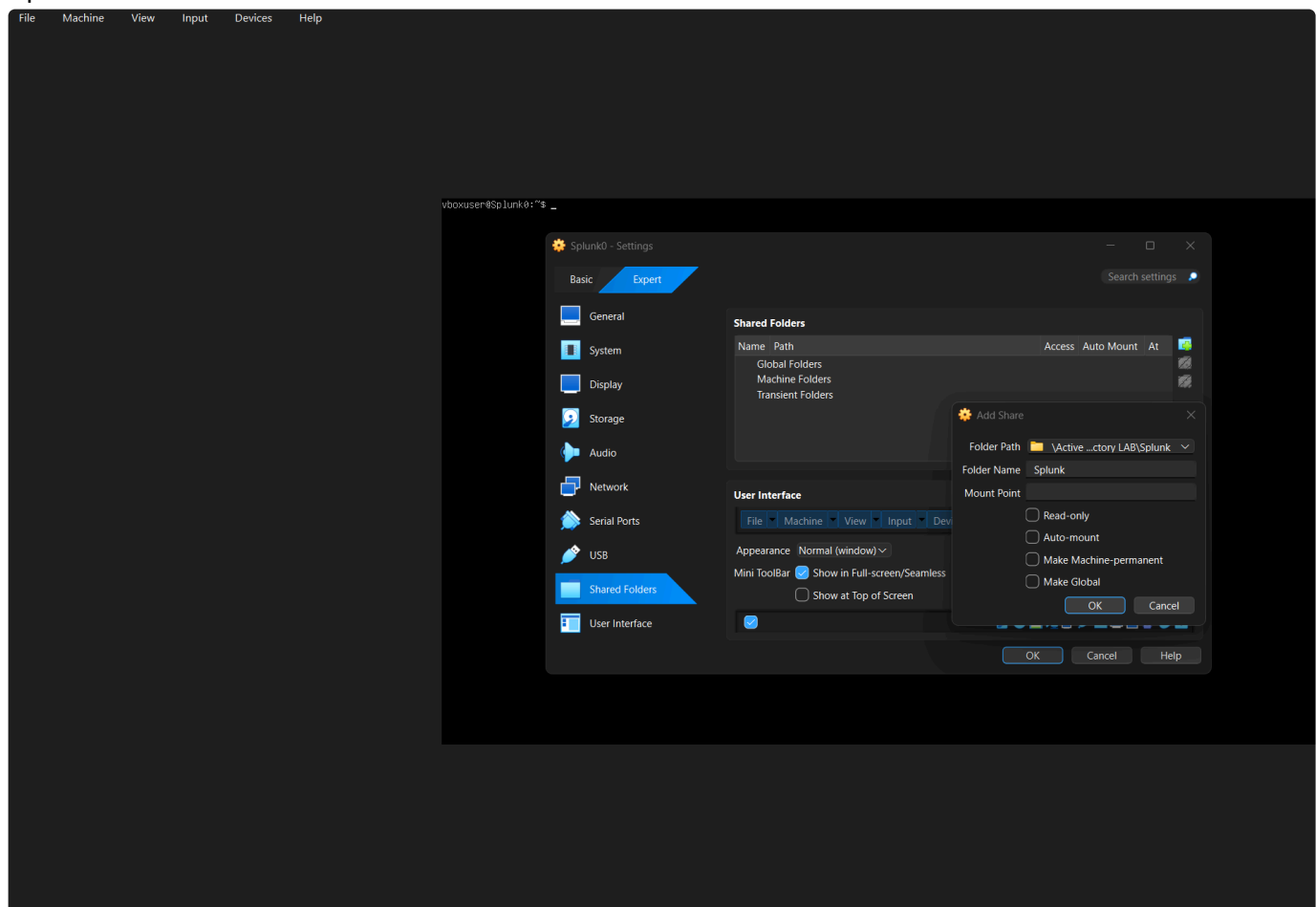
No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
vboxuser@Splunk0:~$
```

On the top left tab, click devices> shared server. For our folder path we want to pick where we put the Splunk deb file



make sure all options are selected

reboot server (sudo reboot) then attempt to add user

Our first attempt at adding the vboxuser to the vboxsf group failed with the error "fatal: The group vboxsf does not exist." This happened because the Guest Additions ISO we installed earlier only makes the installer available, it does not actually install the guest utilities themselves. To fix this we ran "sudo apt-get

install virtualbox-guest-utils", which installs the actual service that creates the vboxsf group and enables shared folder support. Once that finished, the adduser command worked.

```
vboxuser@Splunk0:~$ sudo adduser vboxuser vboxsf
[sudo: authenticate] Password:
fatal: The group `vboxsf' does not exist.
vboxuser@Splunk0:~$ sudo apt-get install virtualbox-guest-utils_
```

(runs now)

```
vboxuser@Splunk0:~$ sudo adduser vboxuser vboxsf
```

Now lets create a new directory called "share" so we can mount our shared folder to share directory

```
vboxuser@Splunk0:~$ mkdir share
vboxuser@Splunk0:~$ sudo mount -t vboxsf -o uid=1000,gid=1000 Splunk share/
```

verify content by changing directory to shares and running ls -la to view all contents of the folder. Once down run "sudo dpkg -i" to install the Splunk Package

```
vboxuser@Splunk0:~/share$ ls -la
total 1293024
drwxrwxrwx 1 vboxuser vboxuser      0 Sep 13 03:14 .
drwxr-x--- 5 vboxuser vboxuser    4096 Sep 13 03:37 ..
-rwxrwxrwx 1 vboxuser vboxuser 1324052468 Sep 13 01:52 splunk-10.4.3-4174a2deda5d-linux-amd64.deb
vboxuser@Splunk0:~/share$ sudo dpkg -i splunk-10.4.3-4174a2deda5d-linux-amd64.deb
vboxuser@Splunk0:~/share$ sudo dpkg -i splunk-10.4.3-4174a2deda5d-linux-amd64.deb
Selecting previously unselected package splunk.
(Reading database... 104284 files and directories currently installed.)
Preparing to unpack splunk-10.4.3-4174a2deda5d-linux-amd64.deb...
Verify that this system has all the commands we will require to perform the preflight step
No need to run the splunk-preinstall upgrade check
Unpacking splunk (10.4.3)...
Setting up splunk (10.4.3)...
find: '/opt/splunk/lib/python3.7/site-packages': No such file or directory
complete
```

```
vboxuser@Splunk0:~/share$ cd /opt/splunk
vboxuser@Splunk0:/opt/splunk$ ls -la
total 5612
drwxr-xr-x 10 splunk splunk    4096 Sep 13 03:49 .
drwxr-xr-x  3 root  root    4096 Sep 13 03:48 ..
-r--r--r--  1 splunk splunk   521 Aug 19 18:13 README-splunk.txt
drwxr-xr-x  3 splunk splunk 12288 Sep 13 03:49 bin
-r--r--r--  1 splunk splunk   57 Aug 19 17:33 copyright.txt
drwxr-xr-x 17 splunk splunk   4096 Sep 13 03:49 etc
-rw-r--r--  1 splunk splunk   427 Sep 13 03:49 ftr
drwxr-xr-x  4 splunk splunk   4096 Sep 13 03:49 include
drwxr-xr-x  9 splunk splunk   4096 Sep 13 03:49 lib
-r--r--r--  1 splunk splunk 59708 Aug 19 18:23 license-eula.txt
drwxr-xr-x  2 splunk splunk   4096 Sep 13 03:49 openssl
drwxr-xr-x  8 splunk splunk   4096 Sep 13 03:49 opt
drwxr-xr-x 20 splunk splunk   4096 Sep 13 03:49 share
-r--r--r--  1 splunk splunk 5622517 Aug 19 18:37 splunk-10.4.3-4174a2deda5d-linux-amd64-manifest
drwxr-xr-x  2 splunk splunk   4096 Sep 13 03:49 swidtag
vboxuser@Splunk0:/opt/splunk$
```

log into splunk user, change into bin directory then run the install "./splunk start". Agree to terms and set up a password

```
vboxuser@Splunk0:/opt/splunk$ sudo -u splunk bash
splunk@Splunk0:~$ cd bin
```

Splunk General Terms (V4 August 2024)

These Splunk General Terms ("General Terms") between Splunk Inc., a Delaware corporation, with its principal place of business at 350 Grandview Street, San Francisco, California 94107, USA ("Splunk" or "we" or "us" or "our") and you ("Customer" or "you" or "your") govern your acquisition, access to, and use of Splunk's Offerings, regardless of how accessed or acquired, whether directly from us or from another approved source. By clicking on the appropriate button, or by downloading, installing, accessing, or using any Offering, you agree to these General Terms. If you are entering into these General Terms on behalf of Customer, you represent that you have the authority to bind Customer. If you do not agree to these General Terms, or if you are not authorized to accept the General Terms on behalf of Customer, do not download, install, access, or use any Offering. The "Effective Date" of these General Terms is: (i) the date of delivery; or (ii) the date you access or use the Offering in any way, whichever is earlier. Capitalized terms are defined in the Definitions section below. Effective September 23, 2024, and unless the context otherwise requires, any reference in these General Terms to "Splunk Inc.", "Splunk", "we", "us" or "our" will be deemed to refer to "Splunk LLC".

1. Your Use Rights and Limits

1.1. Your Use Rights. We grant you a non-exclusive, worldwide, non-transferable and non-sublicensable right, subject to your compliance with these General Terms and payment of applicable fees, to use acquired Offerings only for your Internal Business Purpose during the Term, up to the Capacity, and, if applicable, in accordance with the Order ("Use Rights"). You have the right to make a reasonable number of copies of On-Premises Products for archival and back-up purposes.

1.2. Limits on Your Use Rights. Except as expressly permitted in the Order, these General Terms on Documentation, your Use Rights exclude the right to, and you agree not to (nor allow any user or Third Party Provider to): (i) reverse engineer, decompile, disassemble or otherwise attempt to discover source code or underlying structures, ideas, protocols or algorithms of, or used by, any Offering; (ii) modify, translate or create derivative works based on any Offering; (iii) use an Offering to ingest, process, monitor, analyze or service the devices, systems, networks or application data of any third party; (iv) resell, sublicense, rent the use of, transfer or distribute any Offering; (v) access or use an Offering to analyze, test, characterize, inspect, or monitor its availability, performance, or functionality for competitive purposes; (vi) access or use an Offering to develop, test, troubleshoot, support, or market any software or service that competes with any Offering, or that integrates, interoperates with, or constitutes an extension of any Offering and that you use or intend to use for a commercial purpose; (vii) access or use any Offering in order to analyze, test, characterize, inspect, or monitor its source code or underlying structures, ideas, protocols, or algorithms it contains or uses; (viii) attempt to disable or circumvent any license key or other technological mechanism or measures intended to prevent, limit or control use or copying of, or to protect the intellectual property rights of, Splunk or its licensors.

run this command so that so that Splunk starts up every time our virtual machine reboots

```
vboxuser@Splunk0:/opt/splunk/bin$ sudo ./splunk enable boot-start -user splunk_
```

Install Splunk universal forwarder and Sysmon on Target machine and Server

Splunk Universal Forwarder

The Universal Forwarder is a lightweight piece of software that gets installed on a device (like your Windows 10 client and Active Directory server) to collect logs and forward them to your central Splunk server. It doesn't index or search data itself, it just reads log files and event data locally, then sends that data over the network to Splunk for storage and analysis.

In this lab, the Universal Forwarder is what lets my Windows 10 machine and AD server "talk" to the Splunk server. Without it, Splunk would have no way of seeing what's happening on those machines, since Splunk can only work with data it actually receives.

Sysmon (System Monitor)

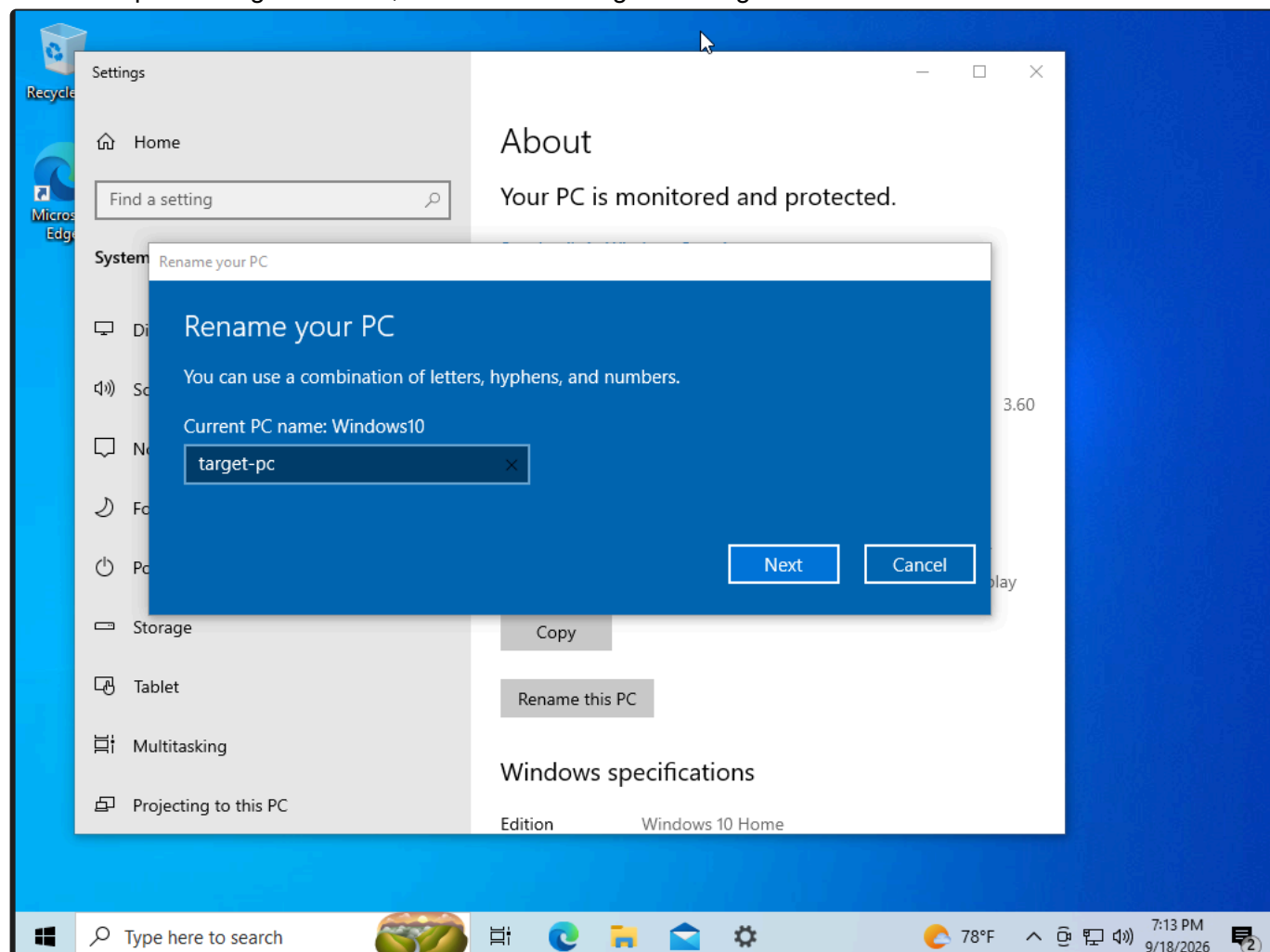
Sysmon is a free Microsoft Sysinternals tool that runs on Windows and logs detailed system activity that Windows doesn't capture by default. This includes things like process creation (what program launched what), network connections, file creation timestamps, and registry changes.

Standard Windows Event Logs are fairly limited out of the box; they'll tell you a process started, for example, but not much detail about it. Sysmon fills that gap by logging much richer, security relevant detail, which makes it possible to actually detect suspicious behavior (like a malicious PowerShell command or an unusual process spawning another process) rather than just seeing surface-level events.

How they work together in this lab:

Sysmon generates the detailed logs on your Windows 11 client, and the Splunk Universal Forwarder is what takes those logs and ships them over to your Splunk server. Once there, Splunk indexes the data so you can search it, build alerts, and create dashboards. This pipeline (Sysmon generates > Forwarder sends > Splunk analyzes) is the backbone of turning raw Windows Server 11 and Windows 11 client activity into something you can actually monitor and investigate, which is the whole point of the blue-team side of this lab.

Before I installed the Universal Forwarder and Sysmon I renamed the Windows machine to "target-pc" by going into This PC properties and restarting the device. I did this so the hostname would clearly identify it as the endpoint being monitored, rather than leaving it as the generic default "Windows10" name.



I noticed that the IP for the target machine(windows 10 VM) was not matching the diagram, the address that was listed would show up under the windows sever. So to fix this issue I went into internet protocol version 4 properties to manually set the IP to 192.168.10.5

(note that this happened because the client was still pulling an address automatically through DHCP instead of using a fixed one)

```
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

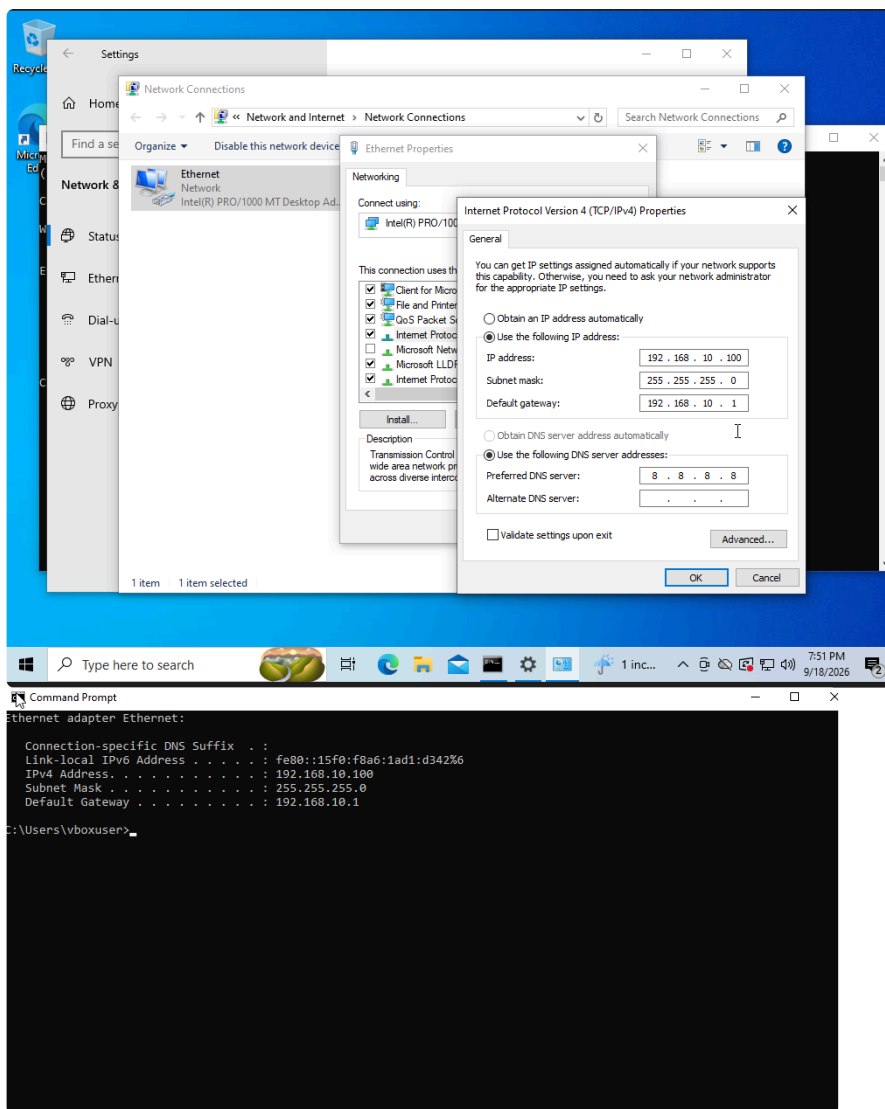
C:\Users\vbouser>ipconfig

Windows IP Configuration

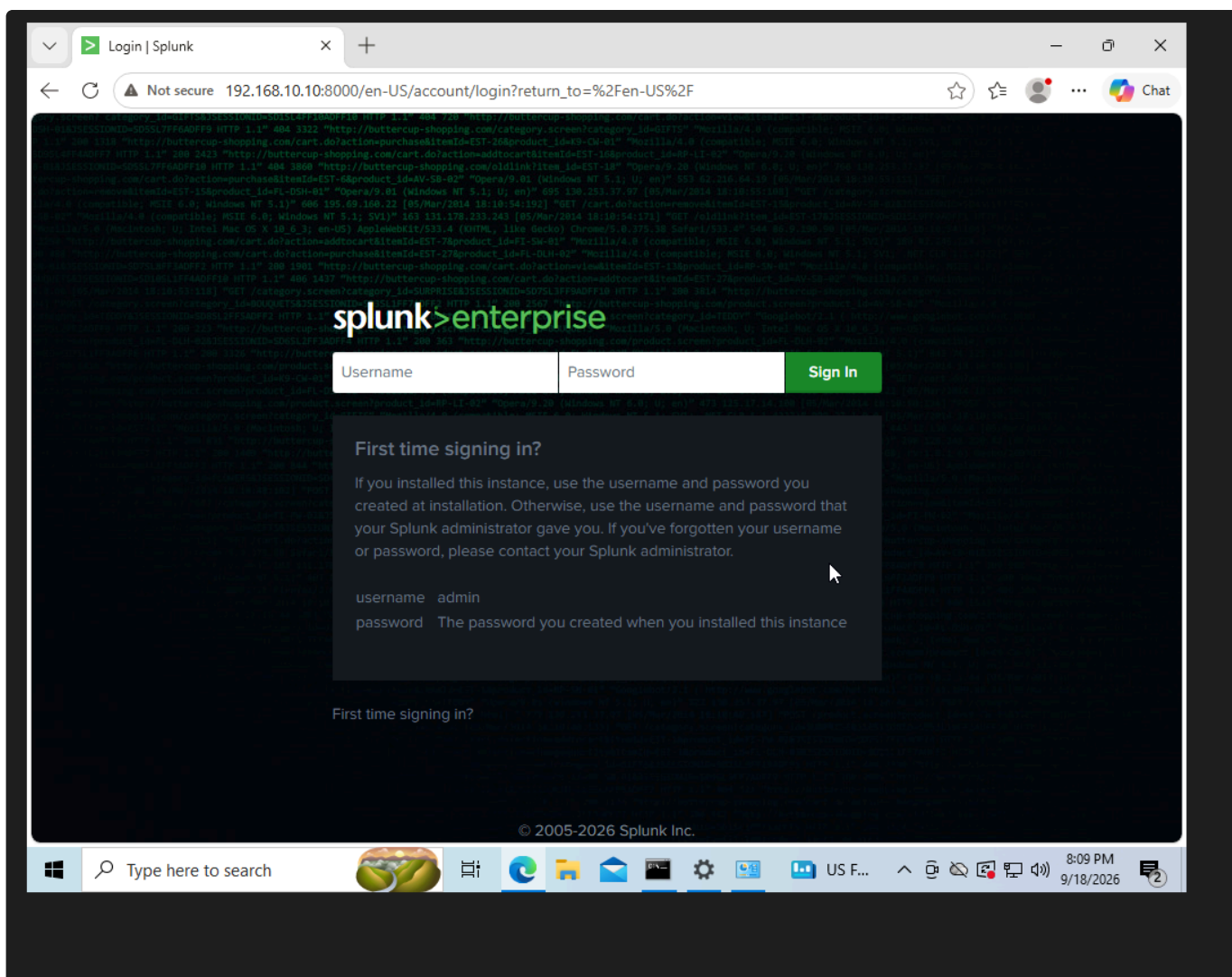
Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : mynetworksettings.com
    Link-local IPv6 Address . . . . . : fe80::15f0:f8a6:1ad1:d342%6
    IPv4 Address. . . . . : 192.168.10.5
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.10.1

C:\Users\vbouser>
```



Once corrected, I head into the splunk server to ensure it can be accessed by launching Microsoft explorer and typing 192.168.10.10:8000



Once all is complete we can now start the task of installing Splunk universal installer

Simply log into the account that was created early, head into free trials and downloads then look for Splunk universal forwarder download the package that matches the system we are using. Our case would be

windows 10 64 bit.

Splunk Universal Forwarder 10.4.3

Universal Forwarders provide reliable, secure data collection from remote sources and forward that data into Splunk software for indexing and consolidation. They can scale to tens of thousands of remote systems, collecting terabytes of data.

Choose Your Installation Package

Windows Linux Mac OS Free BSD Solaris AIX

32-bit **Windows 10** .msi 64.97 MB **Download Now** [Copy wget link](#) More ▾

64-bit **Windows 10, 11 Windows Server 2019, 2022, 2025** .msi 152.05 MB **Download Now**

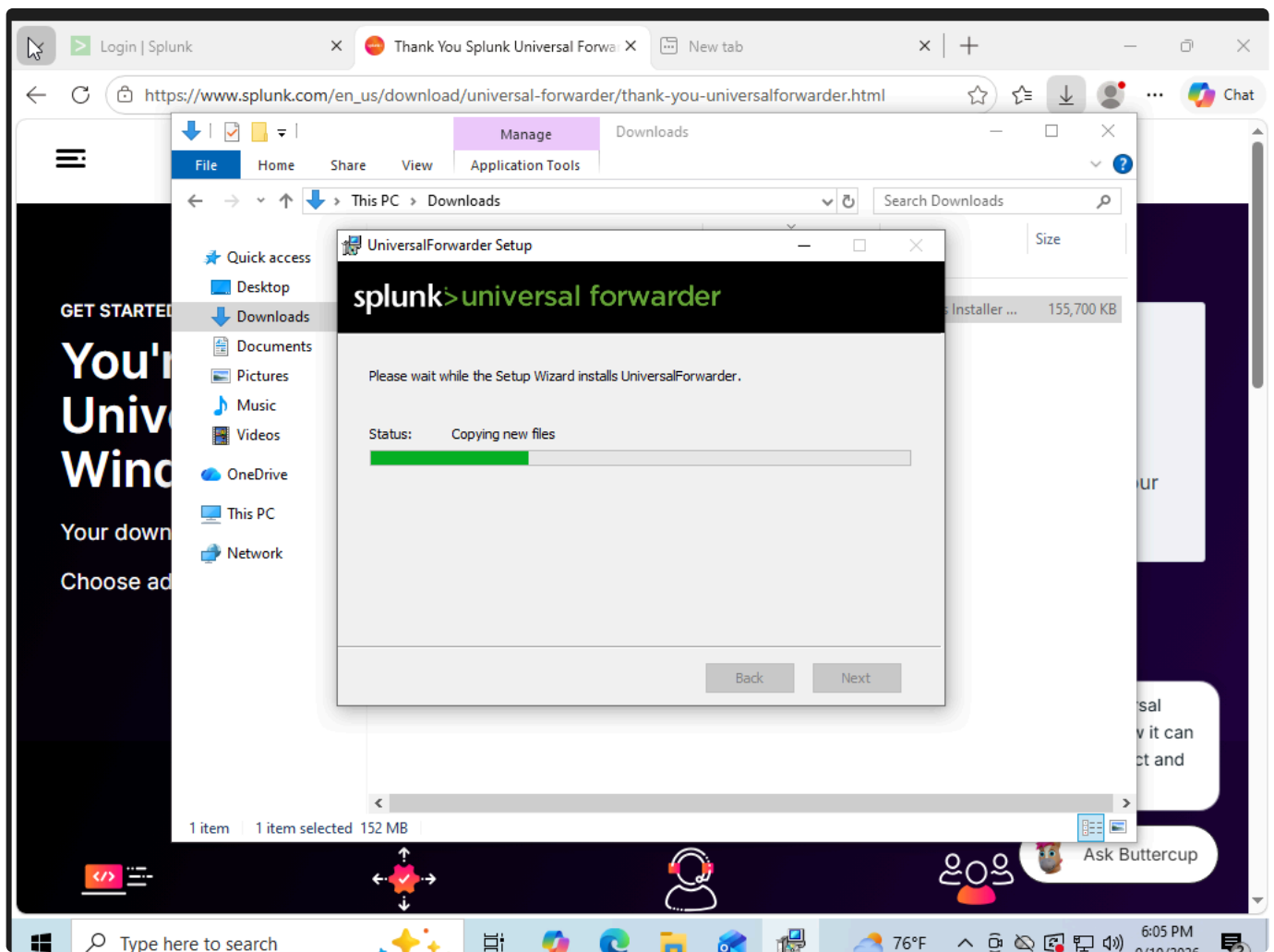
I see you're checking out the Universal Forwarder. Are you interested in how it can scale across your platforms to collect and send data efficiently?

[Release Notes](#) | [System Requirements](#) | [Previous Releases](#) [Ask Buttercup](#)

Agree to terms while in setup and select the on-premises Splunk enterprise instance

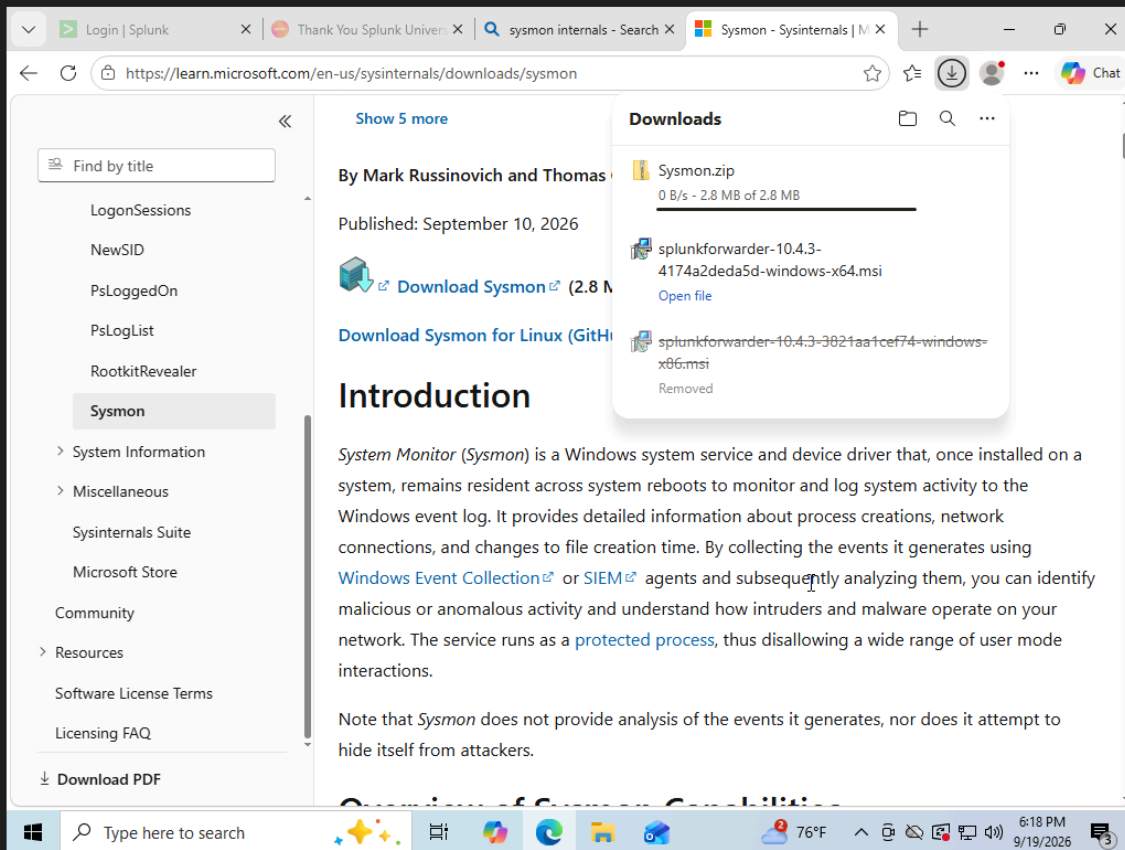
I set username to "Admin" and let it generate a password

Skip the deployment server since we don't have one but for receiving indexer will be 192.168.10.10 with the port 9997 (default port when receiving events)

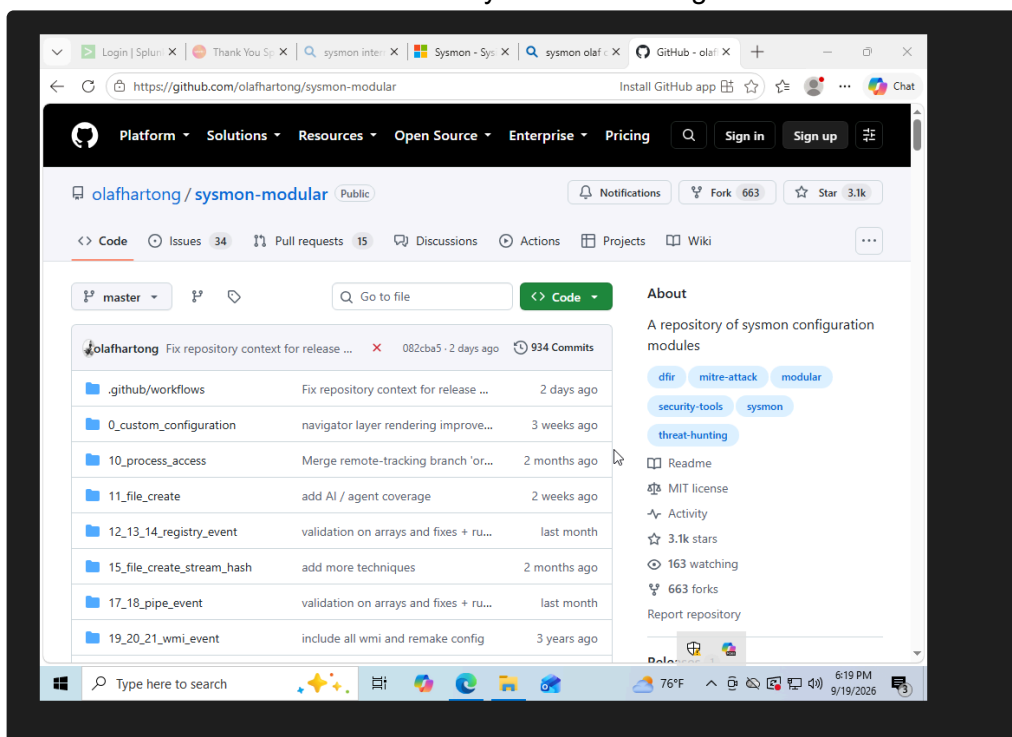


Installing and configuring Sysmon

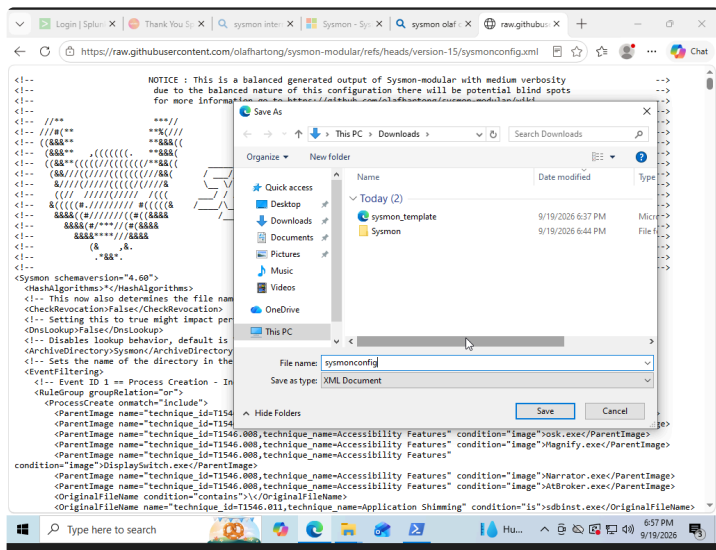
While installing we will download Sysmon search up Sysmon internals and you find a link by Microsoft, download this



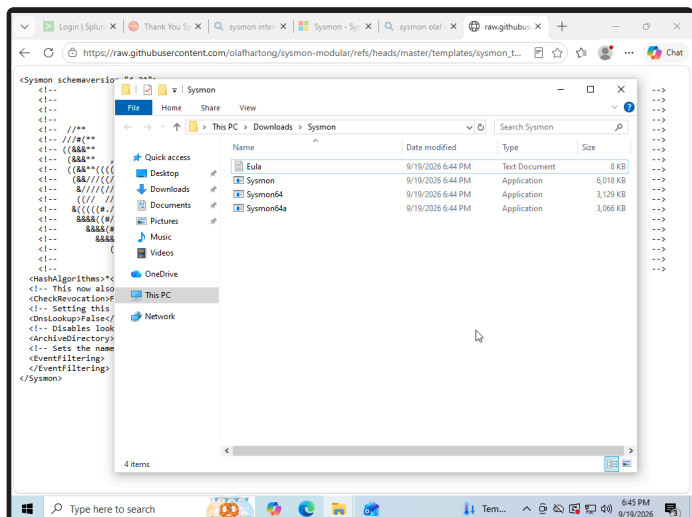
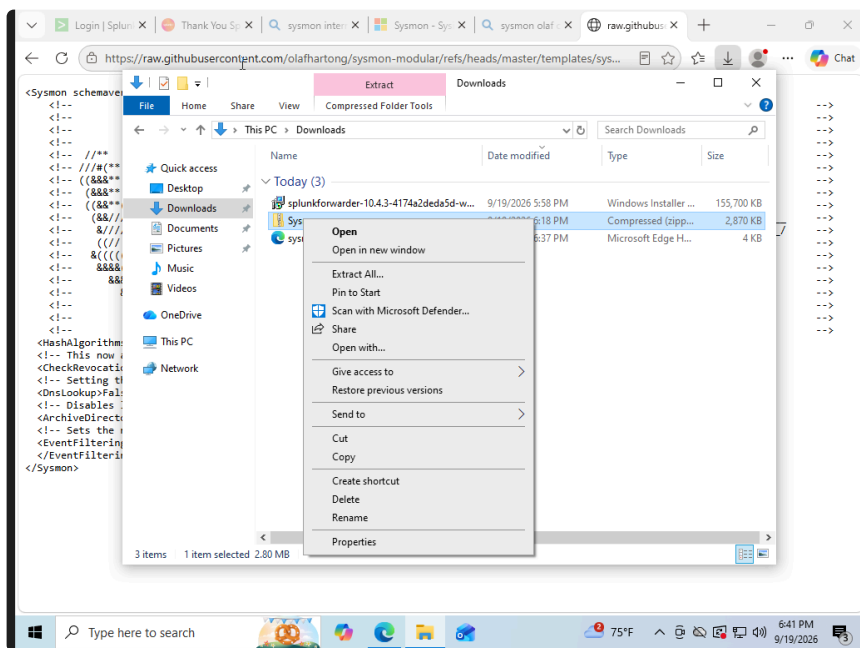
once downloaded we will search for sysmon olaf config on GitHub



search for the Sysmon config (note that i went to version 15 to find), once found select raw and right click. You want to click save as then move the file under the downloads directory

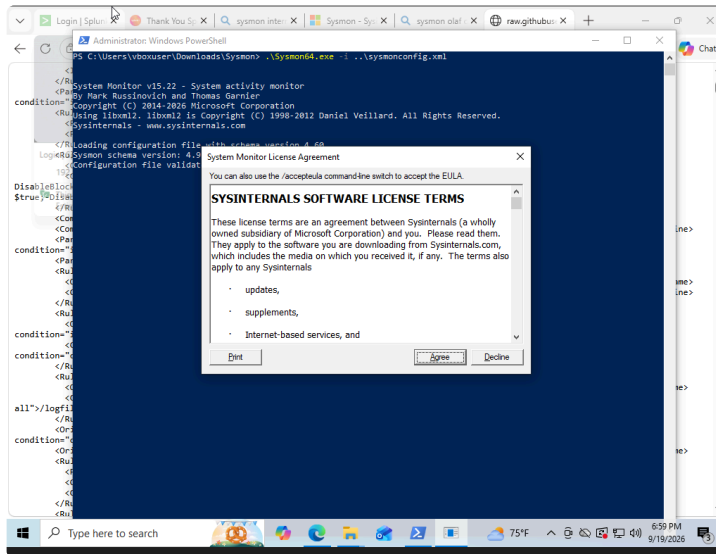


go back to file explorer and extract all from the Sysmon file that we downloaded earlier



copy the path address then go to power shell as an admin and we will change the directory of the file path. We will use -i to specify that we will use a configuration file

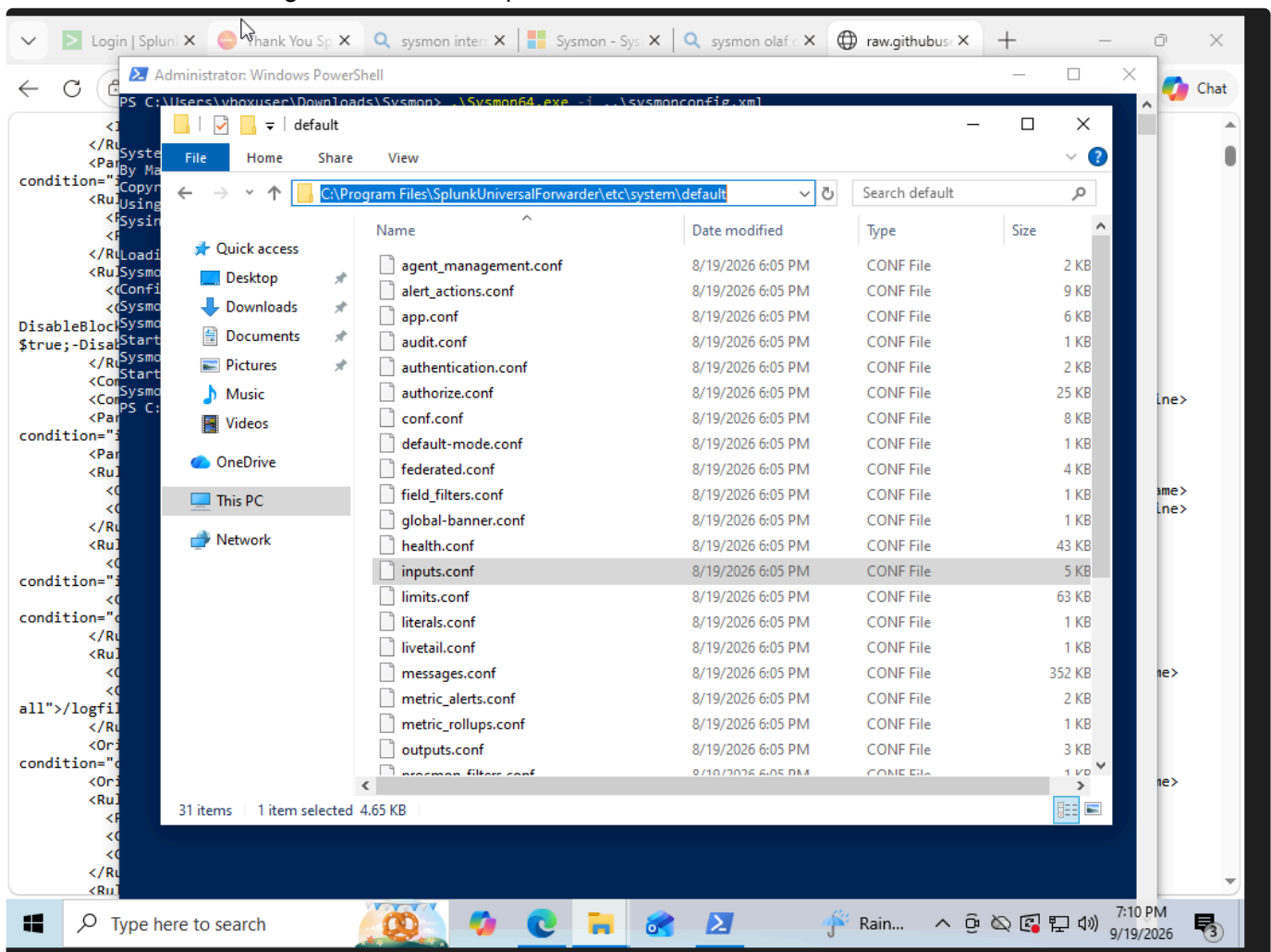
note that we will use ..\ since the template is located in downloads directory



Configuring the forwarder to send data

Now we need to instruct our Splunk forwarder on what we want to send over to our Splunk sever

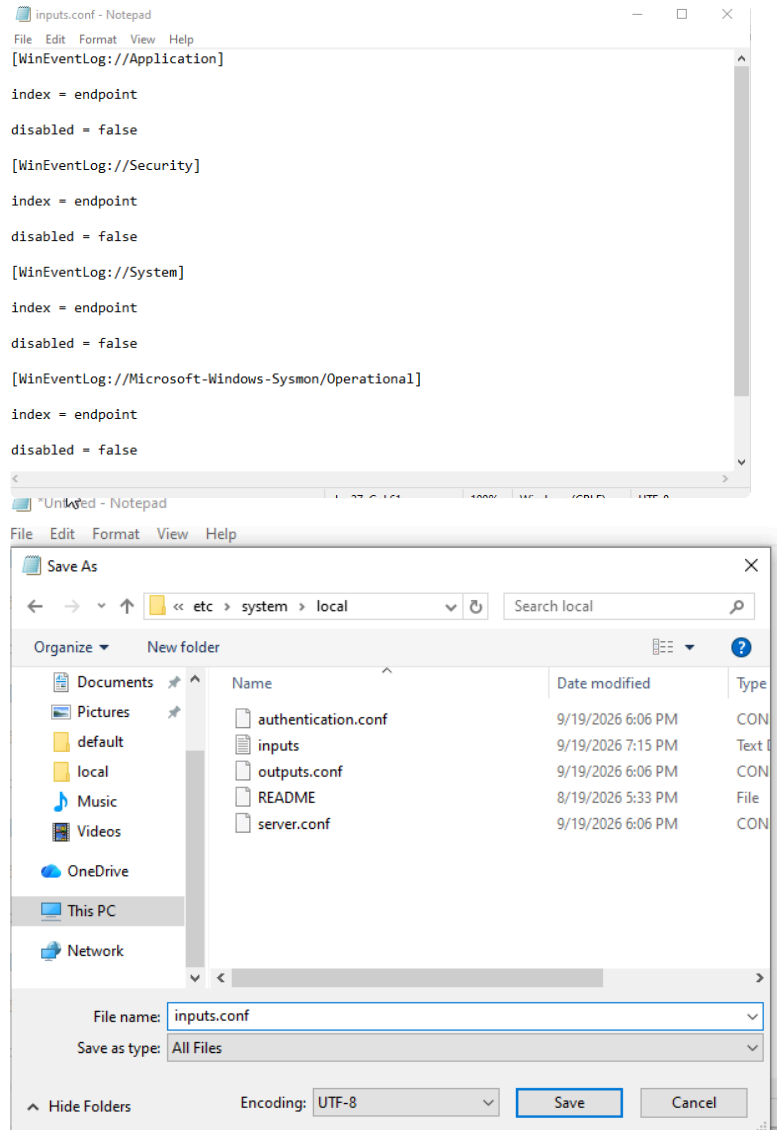
to do this we must configure a file called inputs.conf



copy it and create a new under the local directory (note you don't ever want to edit files found in the default directory, there are supposed to be there incase you mess up and to replace)

go to note pad and create a new file running as administrator. Add this content to the notepad which is instructing our Splunk forwarder to push events related to applications, security, and systems and Sysmon over to our Splunk server

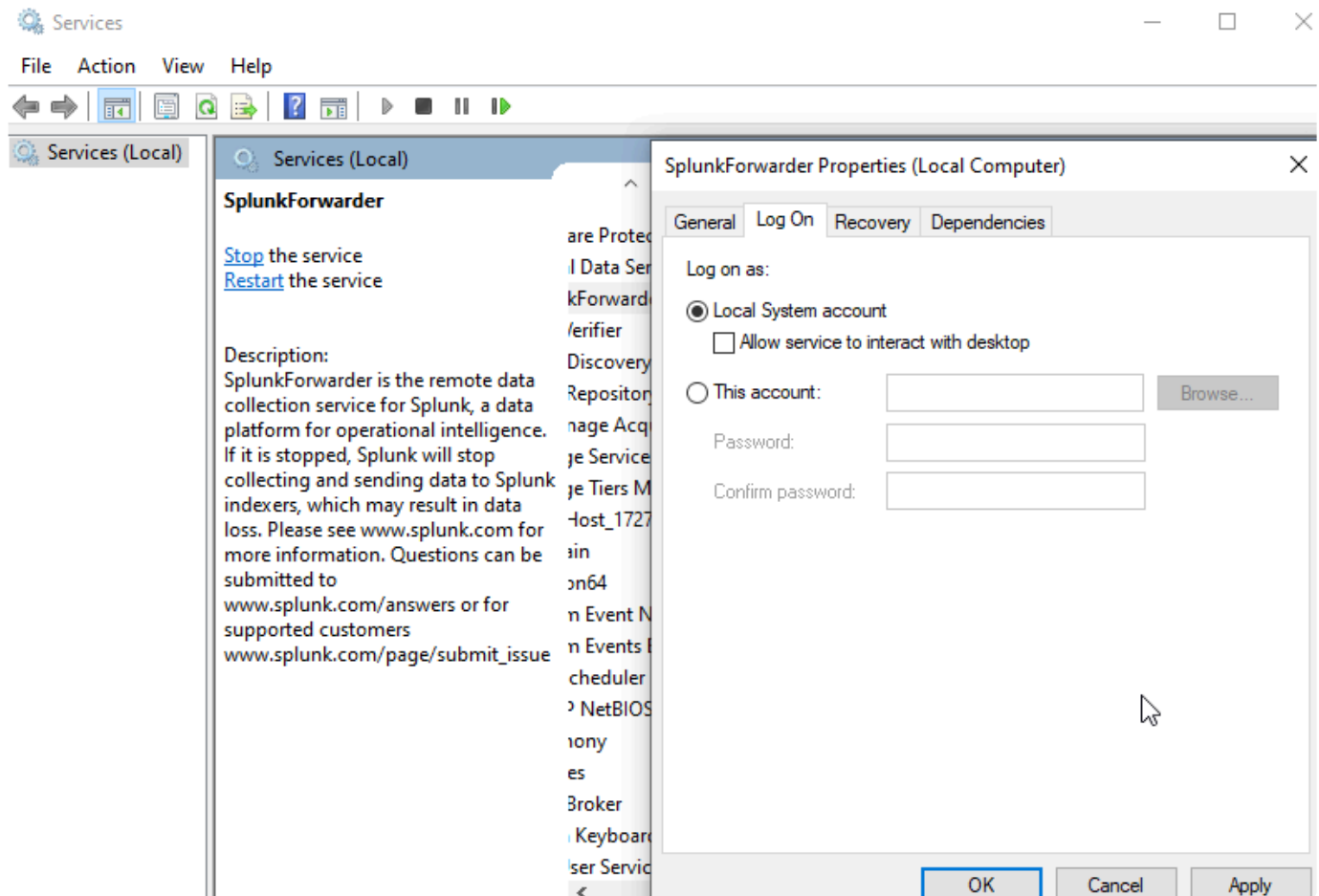
Save file as inputs.conf and type as all files under the local directory within the splunk universal folder



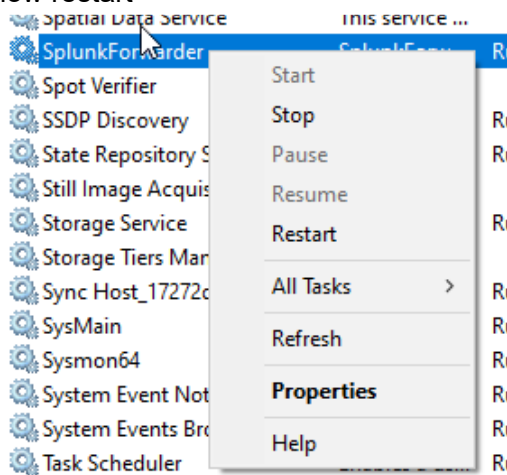
Note whenever you update your inputs.conf file you must restart splunks universal service

To do so search for the services application and run as an administrator. Look for Splunk forwarder

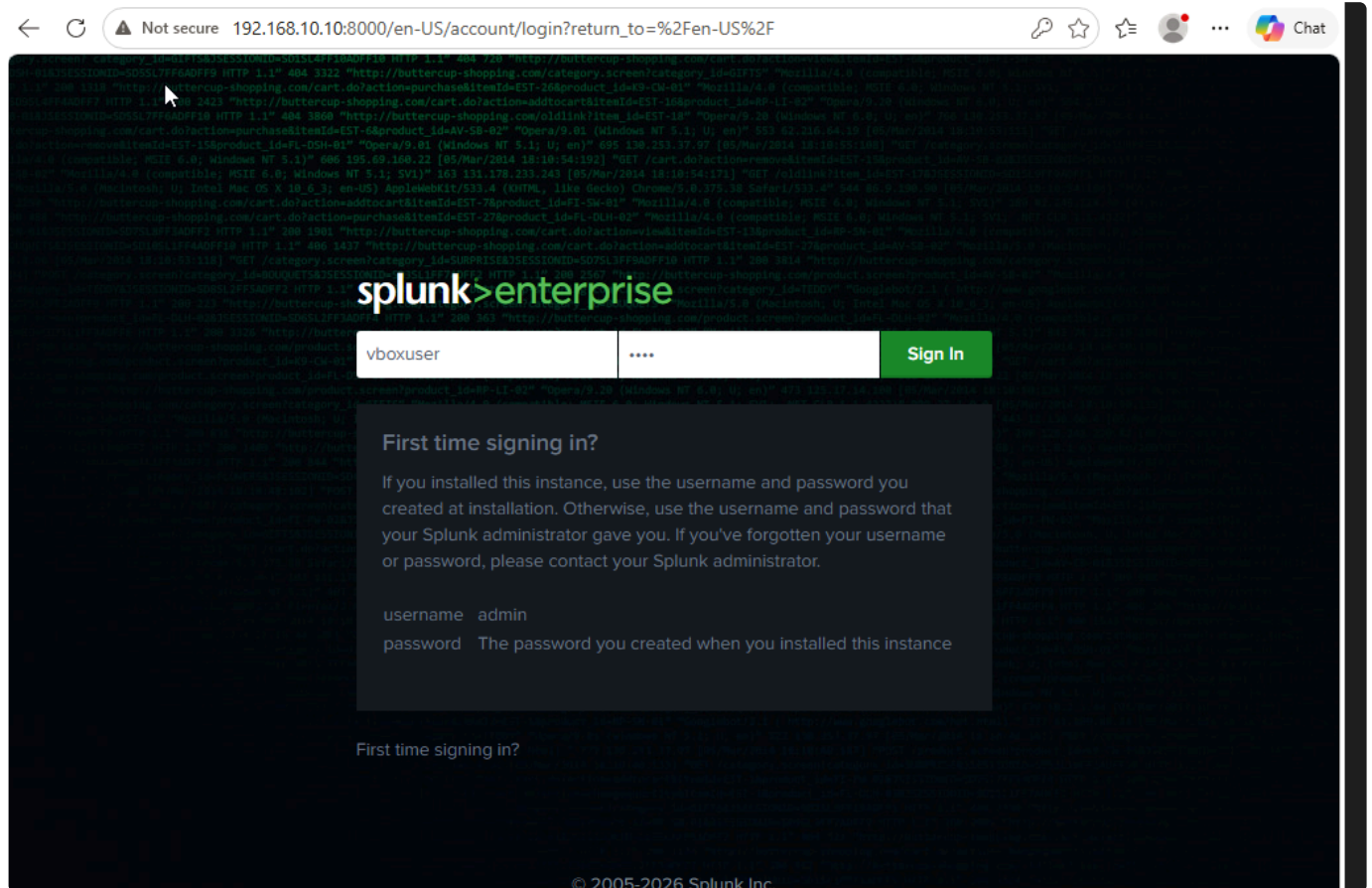
(note while in systems properties, change log on to local system account before we restart)



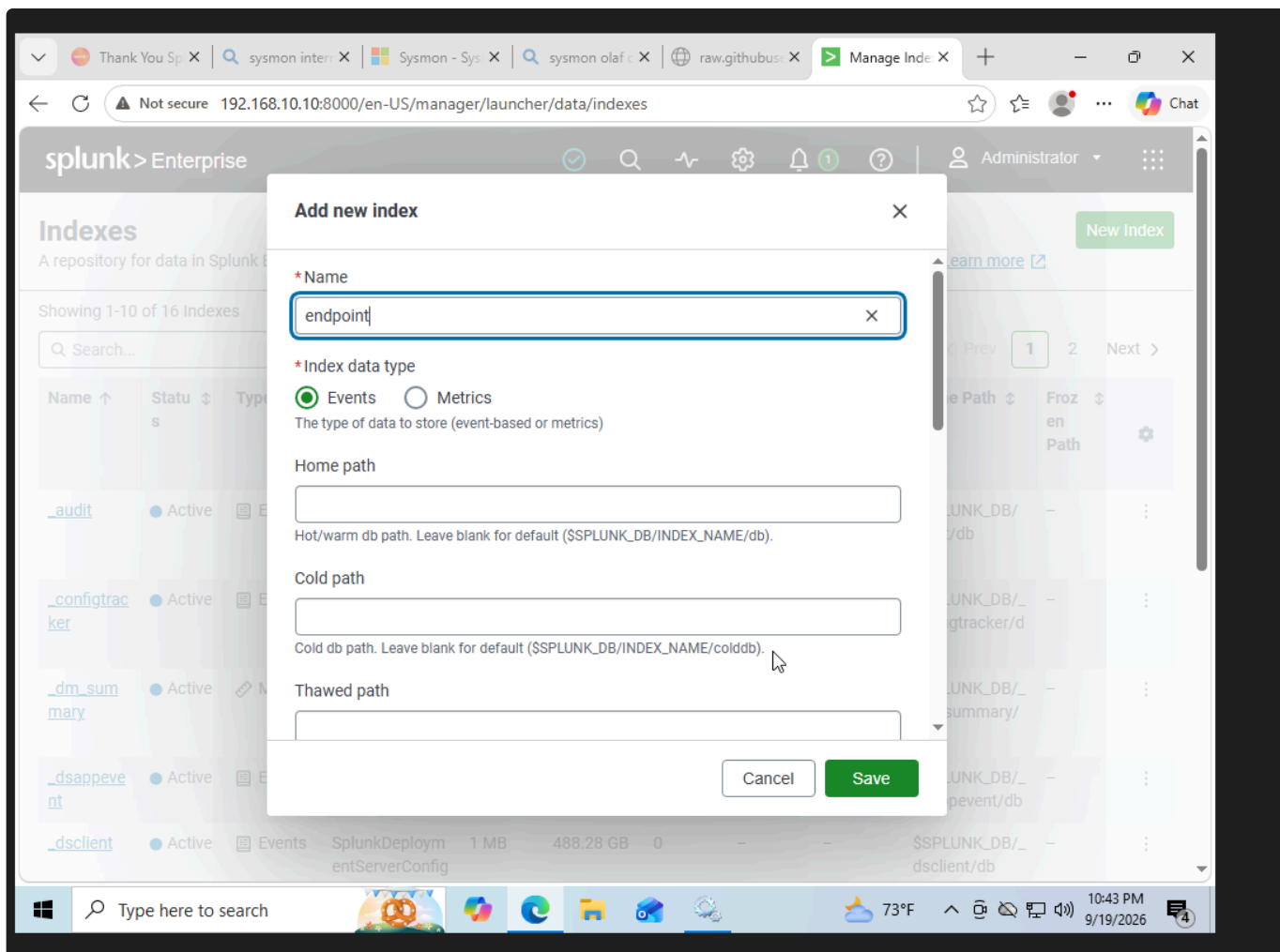
now restart



Lets log on now



When I went to Settings > Indexes, I noticed we didn't have an index called "endpoint," so I created one. We need this because all of our events in inputs.conf are being routed to an index named endpoint, and without that index existing, the data has nowhere to go.



Now head to settings > forward and receiving and add a new receiving port under the configure receiving section and add the port 9997

Now head to apps > search and report and type "Index=endpoint"

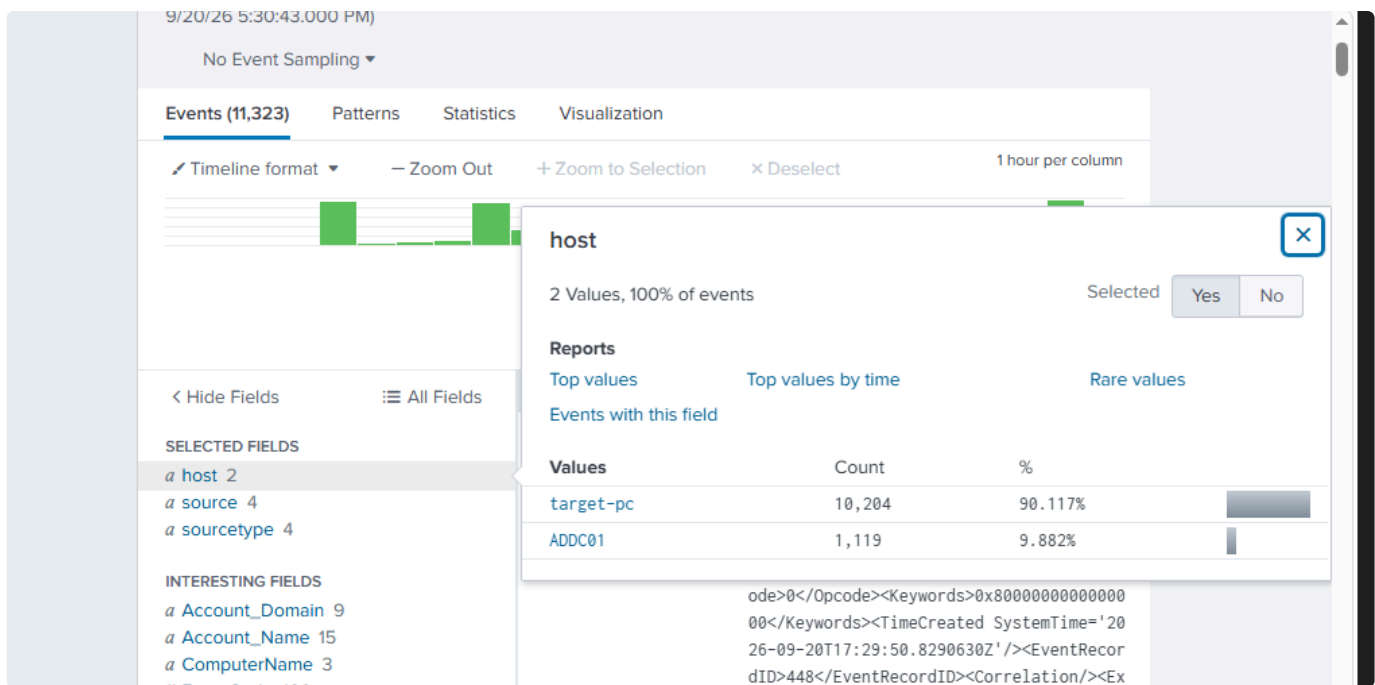
The screenshot shows the Splunk search interface. The search bar contains the query `index="endpoint"`. The results show 7,117 events from 9/19/26 3:00:00.000 AM to 9/20/26 3:19:52.000 AM. The timeline visualization shows a single event at 3:19:46.059 AM on 9/20/26. The event details are as follows:

Time	Event
9/20/26 3:19:46.059 AM	<pre><Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698ffbd9}' /><EventID>11</EventID><Version>2</Version><Level>4</Level><Task>11</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-09-20T03:19:46.0597615Z' /><EventRecordID>3617</EventRecordID><Correlation><Execution ProcessID='2684' ThreadID='3456' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>target-pc</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'>technique_id=T1574.010,technique_name=Services File Permissions Weakness</Data><Data Name='UtcTime'>2026-09-20 03:19:46.059</Data><Data Name='ProcessGuid'>{2fd696e8-3c27-6aaf-2100-000000000000}</Data><Data Name='ProcessId'>1332</Data><Data Name='Image'>C:\Windows\system32\svchost.exe</Data></EventData></Event></pre>

the search returned 7,117 events over the last 24 hours, confirming the pipeline was fully functional end to end. The events show data arriving from the host target-pc, which is my Windows 10 client, with logs coming from multiple sources including Sysmon, Security, and System event channels. This confirms Sysmon is successfully generating detailed activity logs, the Universal Forwarder is capturing them, and Splunk is correctly indexing and displaying that data for searching. With this pipeline verified as working, I can now move forward with building searches, dashboards, and detections using this data

REPEAT STEPS ON THE WINDOWS 2022 SERVER

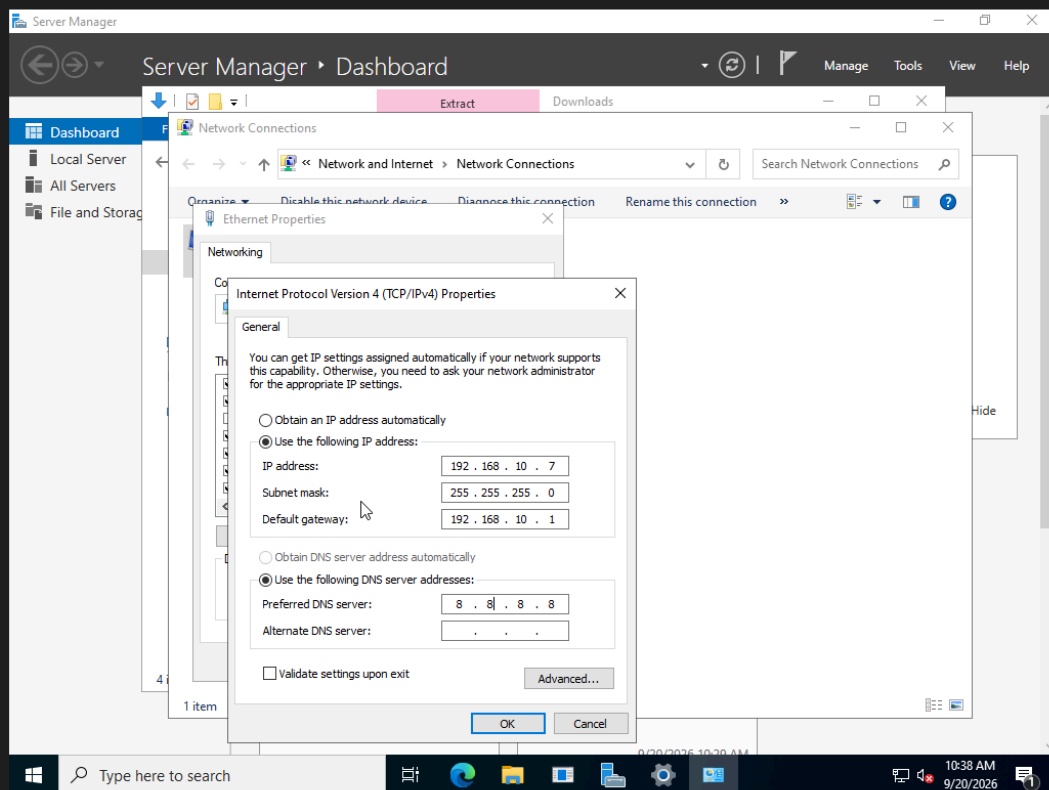
Now we can see 2 host



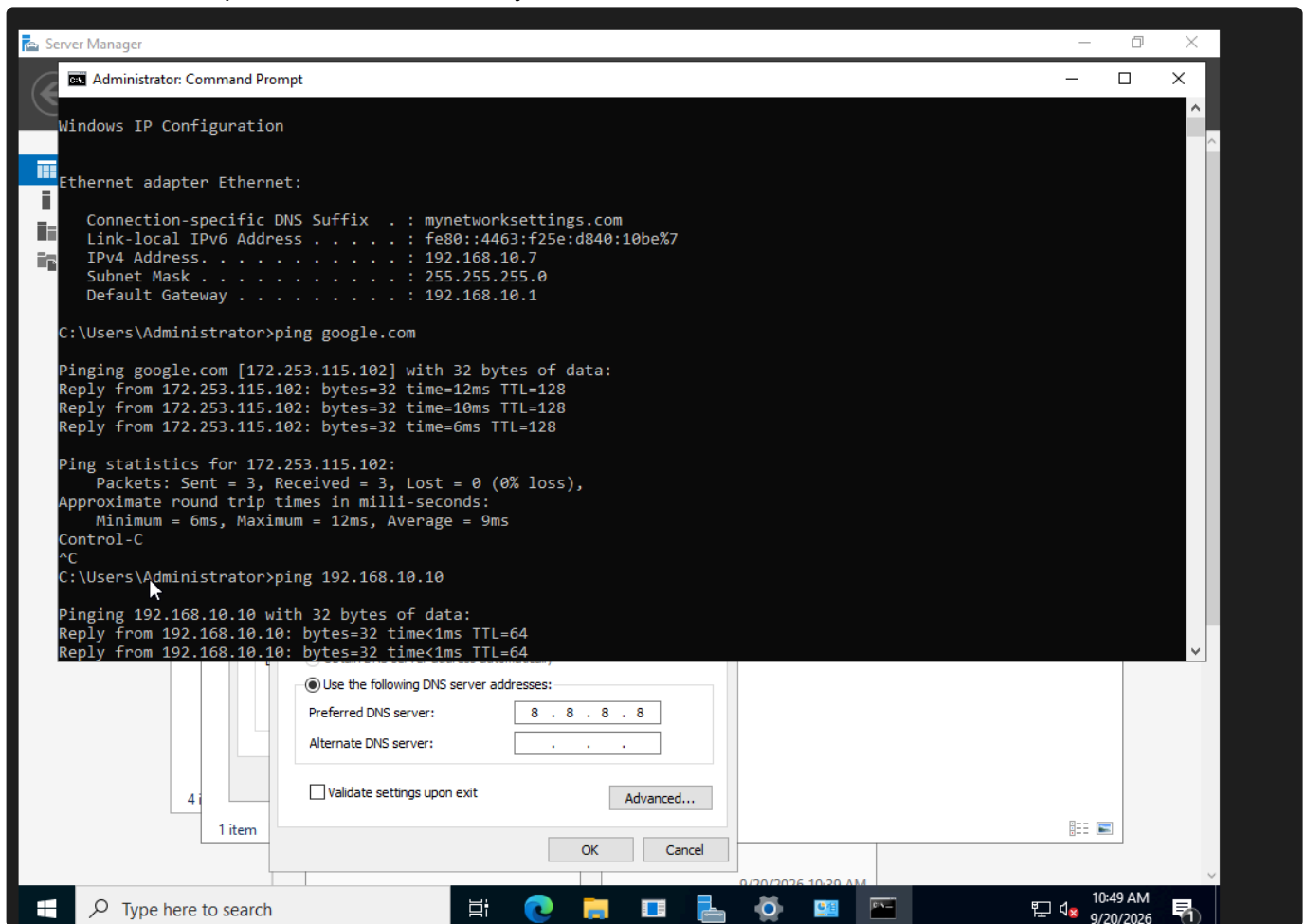
Installing & Configuring Active directory to our Windows Server | Promote Domain Controller | Connect target PC to domain

Installing & Configuring Active directory to our Windows Server

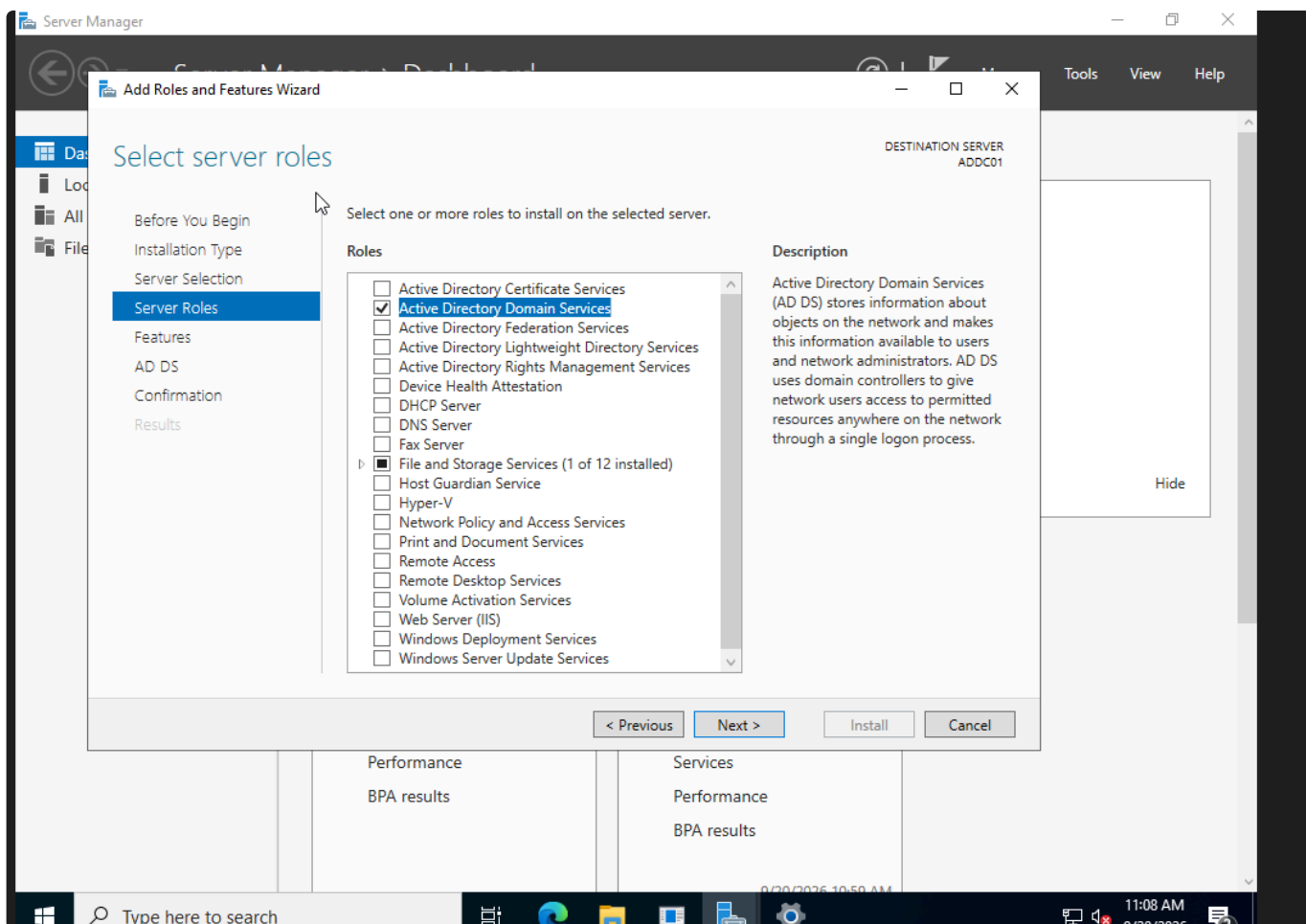
Start by assigning the windows 2022 server a static IP which matches our diagram. To do so head into Internet Protocol Version 4 properties which can be found in ethernet properties



Run ipconfig to confirm IP changes then ping both google and our Splunk machine to ensure communication is possible and connectivity is active.



Head to server manager click manage on the top right > add roles and features > click role based installation > click Active directory domain services the proceed with default settings

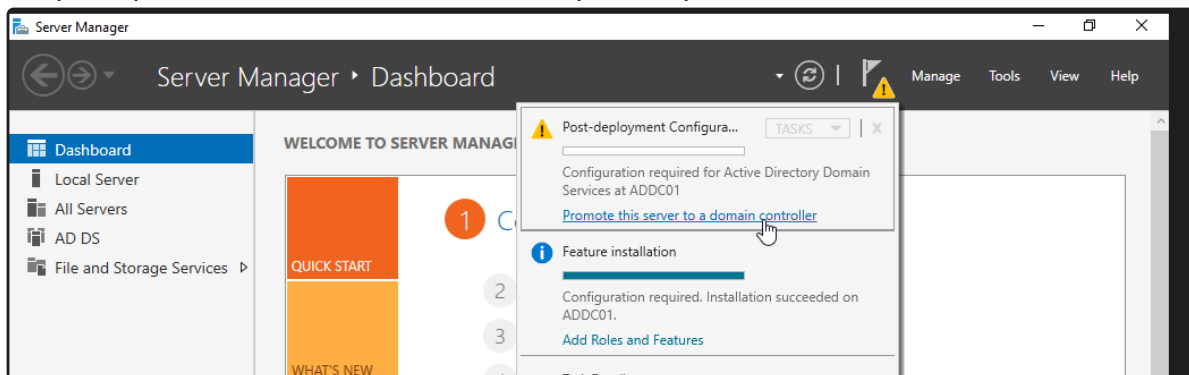


Once done with installation select promote this server. Add a new forest and when specifying name ensure its has "----" for my case i used vbox.local

In the domain controller option section leave default settings and create a new password.

In path leave it as default. Keep a heads up that hackers like to target domain controllers since they carry access to everything and contains password hashes.

run prerequisite check then install. once setup is completed it will restart



Deployment Configuration

TARGET SERVER
ADDC01

Deployment Configuration

Domain Controller Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select the deployment operation

- ☐ Add a domain controller to an existing domain
- ☐ Add a new domain to an existing forest
- ☒ Add a new forest

Specify the domain information for this operation

Root domain name:

vbox.local

Paths

TARGET SERVER
ADDC01

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Specify the location of the AD DS database, log files, and SYSVOL

Database folder:

C:\Windows\NTDS

...

Log files folder:

C:\Windows\NTDS

...

SYSVOL folder:

C:\Windows\SYSVOL

...

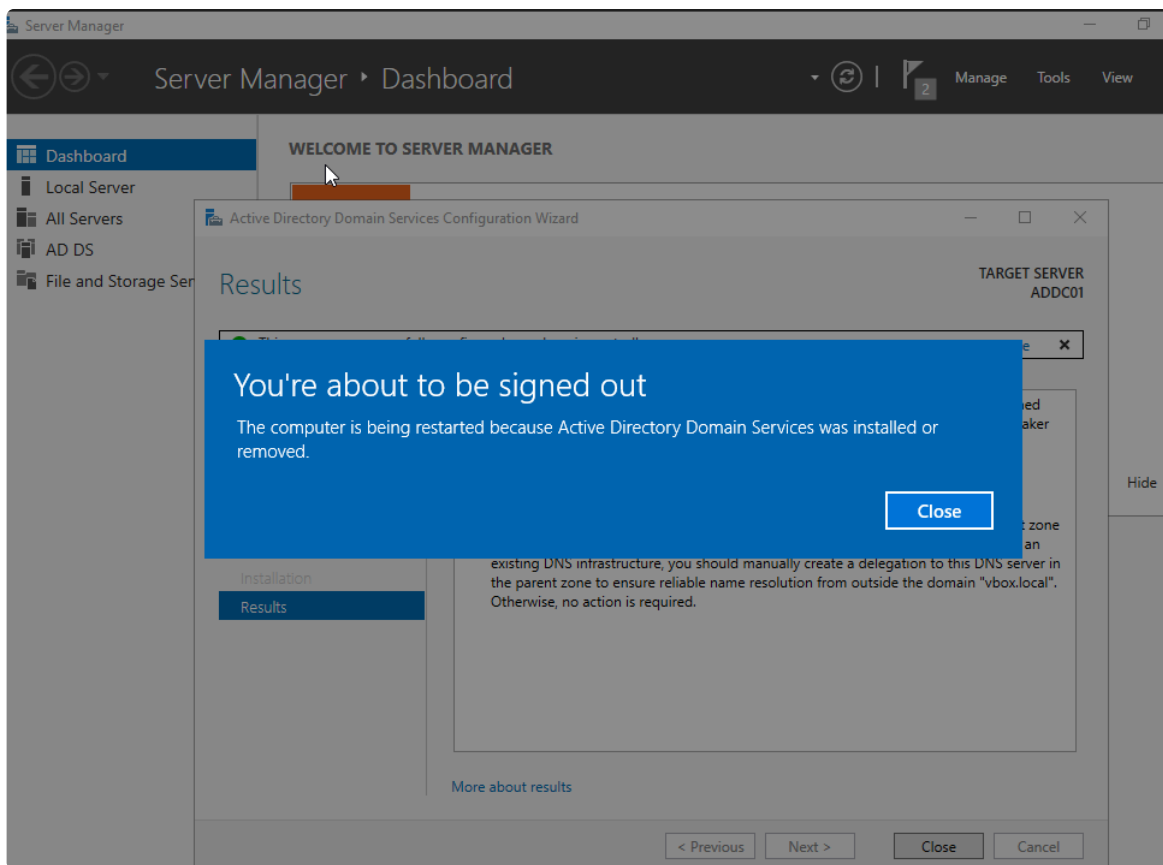
[More about Active Directory paths](#)

< Previous

Next >

Install

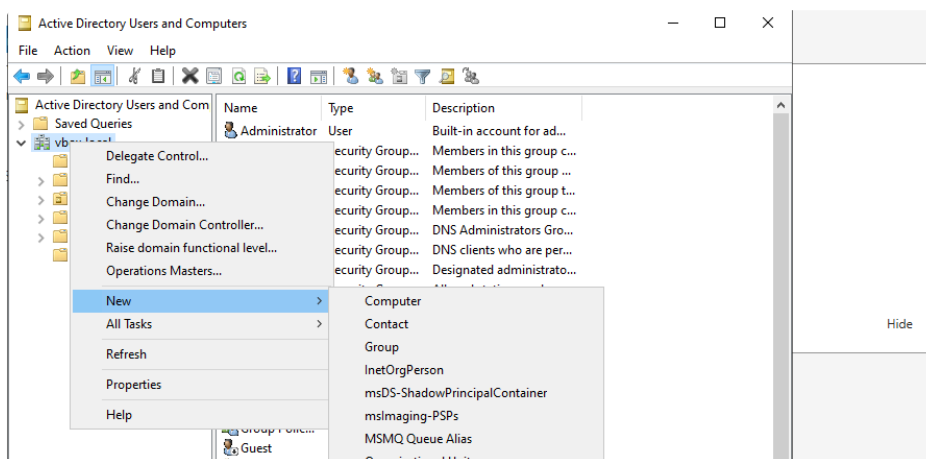
Cancel

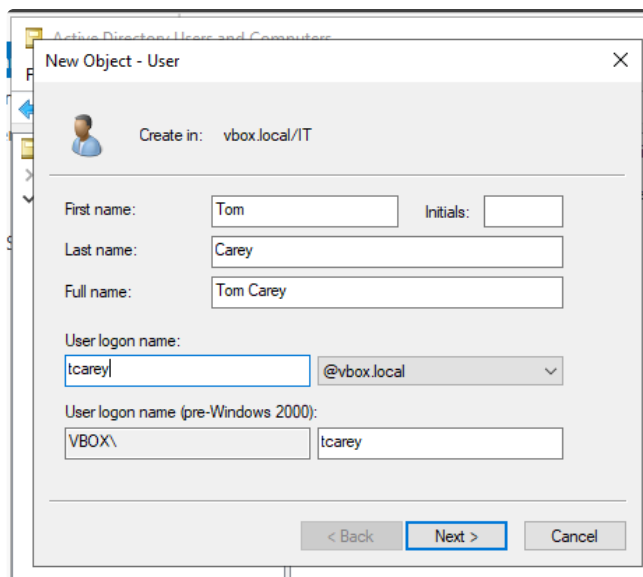


Promote Domain Controller

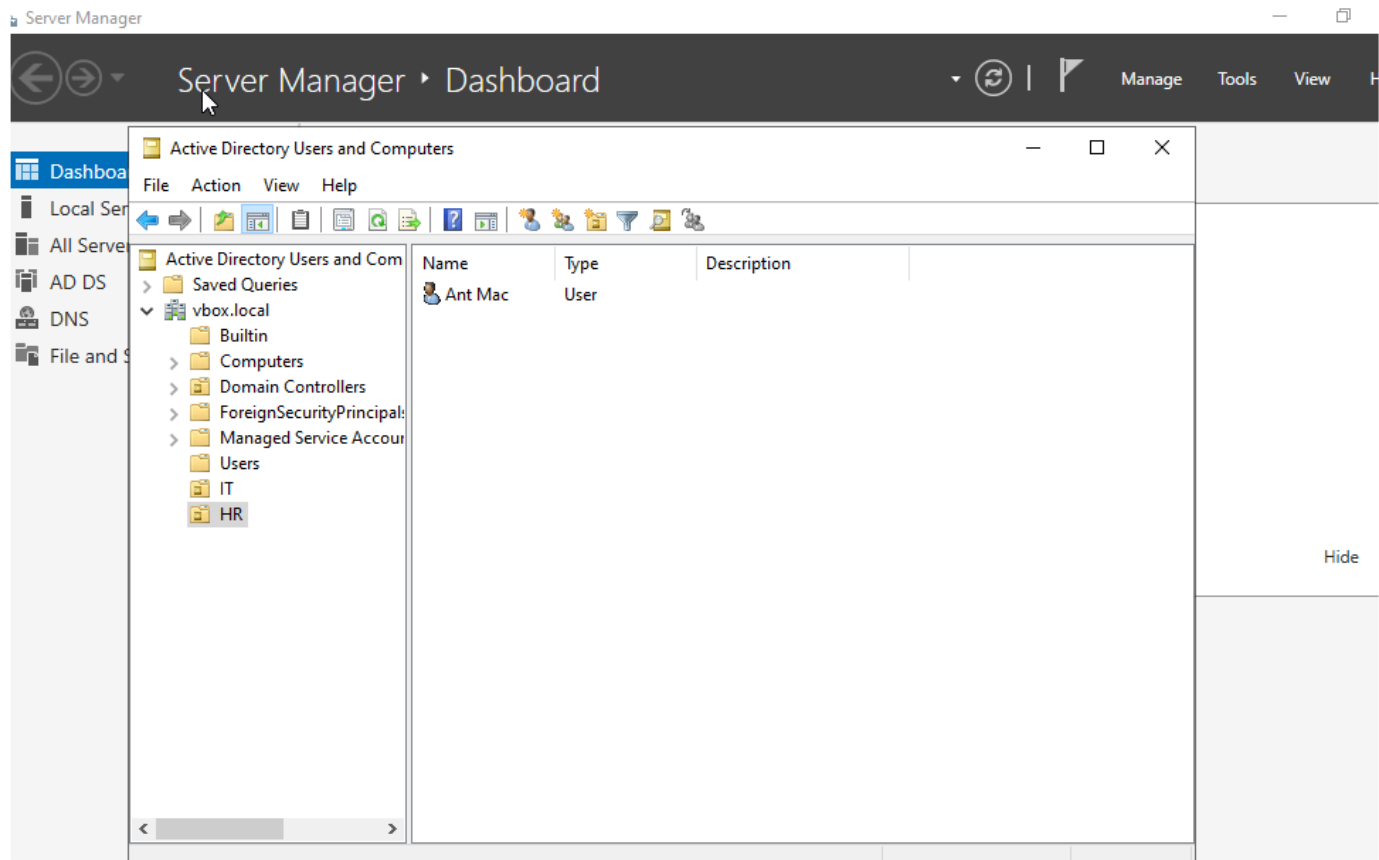
Head to tools > Active directory users and computers. Right click the domain and set up a new organizational unit called "IT"

Lets create a new user inside that organization unit named Tom Carey and set up a password





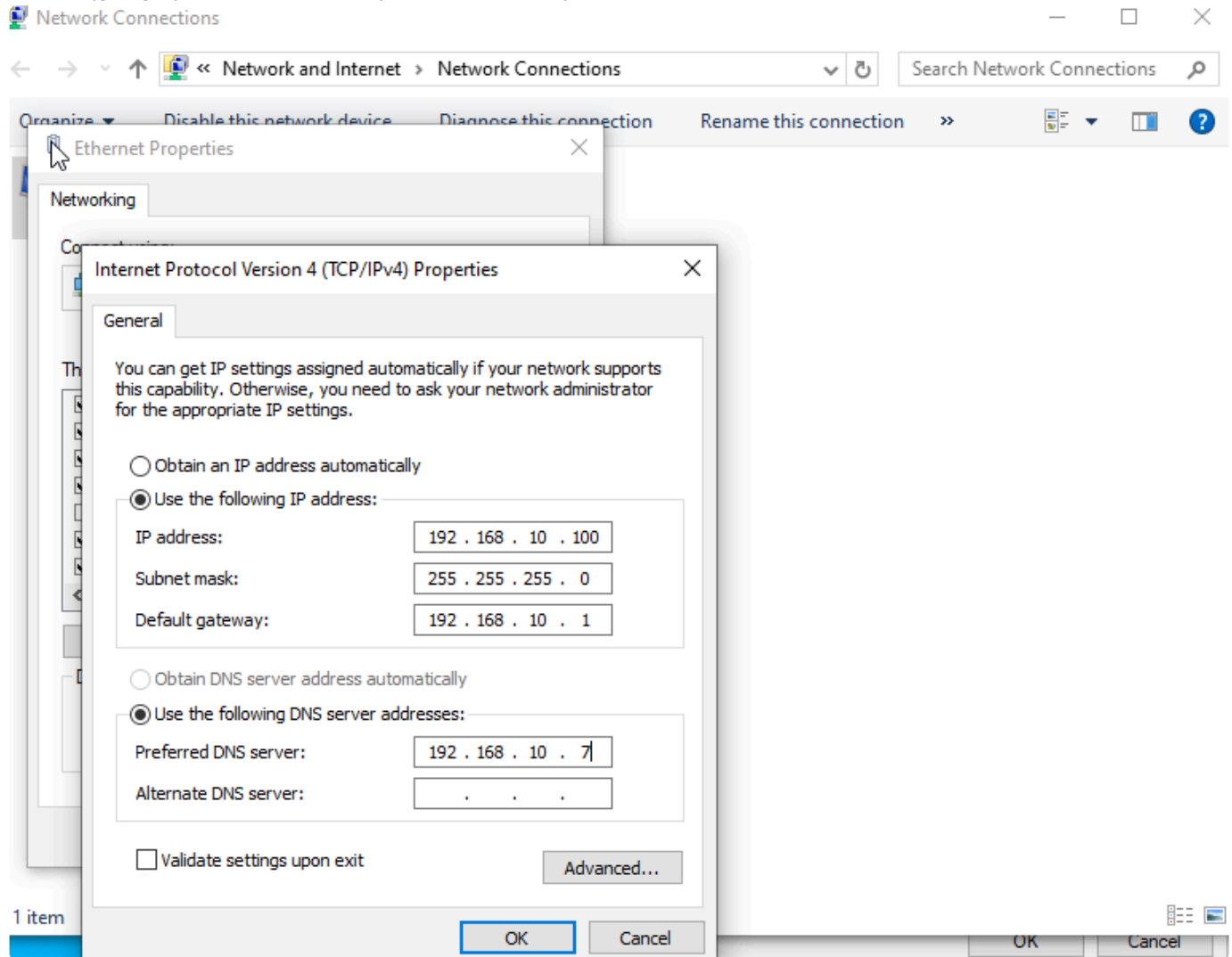
Lets create a new OU called HR and add new user



Connect target PC to domain

Open the target server and change the ipv4 properties by channing the preferred dns server from

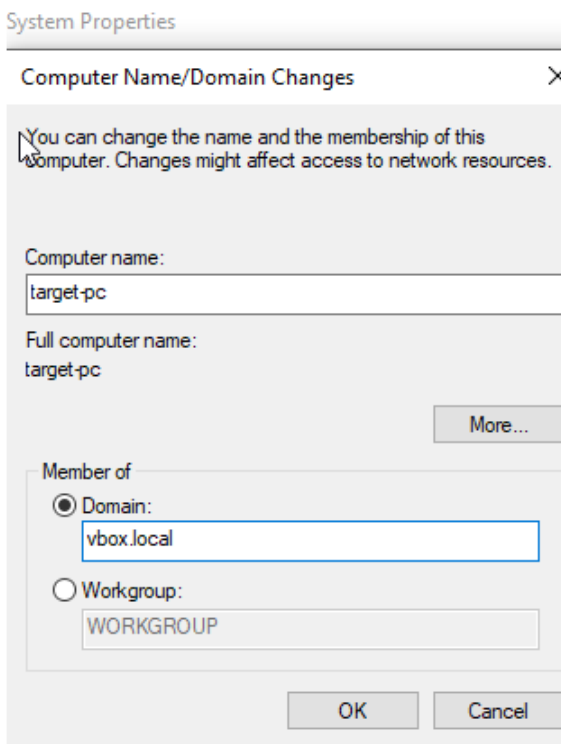
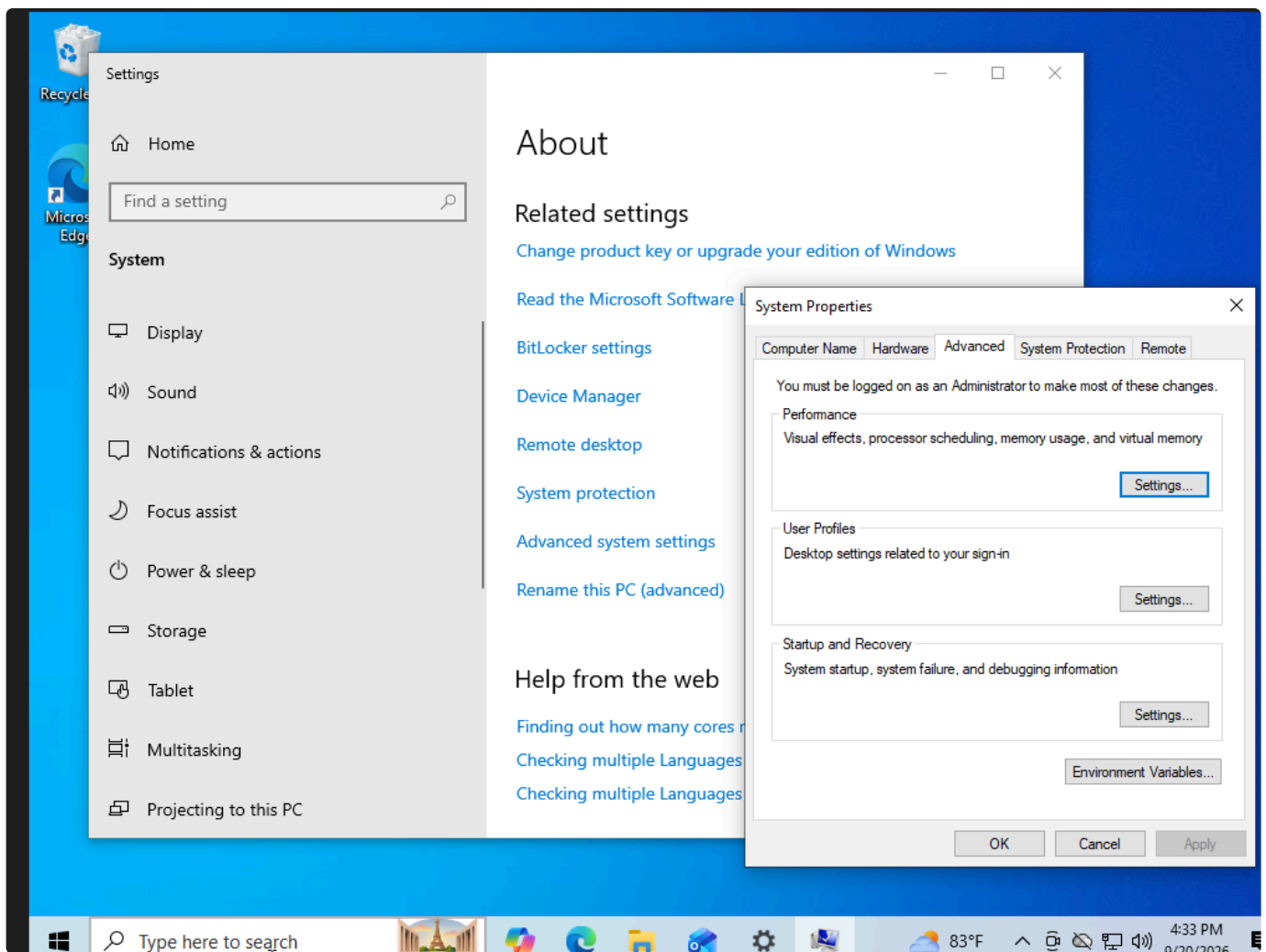
8.8.8.8(google) to 192.168.10.7 (our 2022 server)



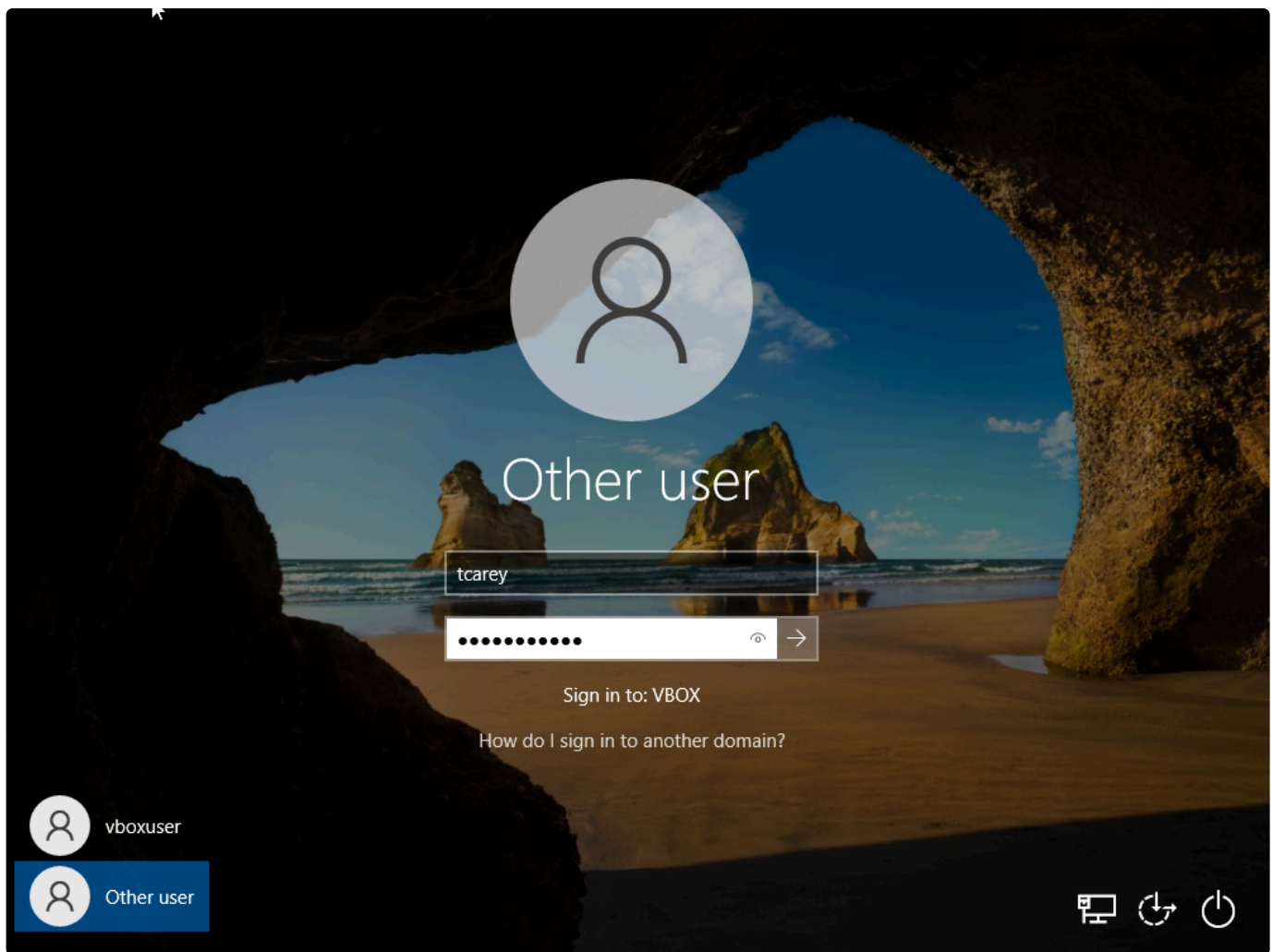
Next go to settings > about > advance system settings > computer name > change and now lets change it to our domain from earlier.

Use credentials that you had created for the 2022 server. restart the pc to apply changes

In a real world environment you would create users and put them into a custom group that are authorized to allow users to join the domain.



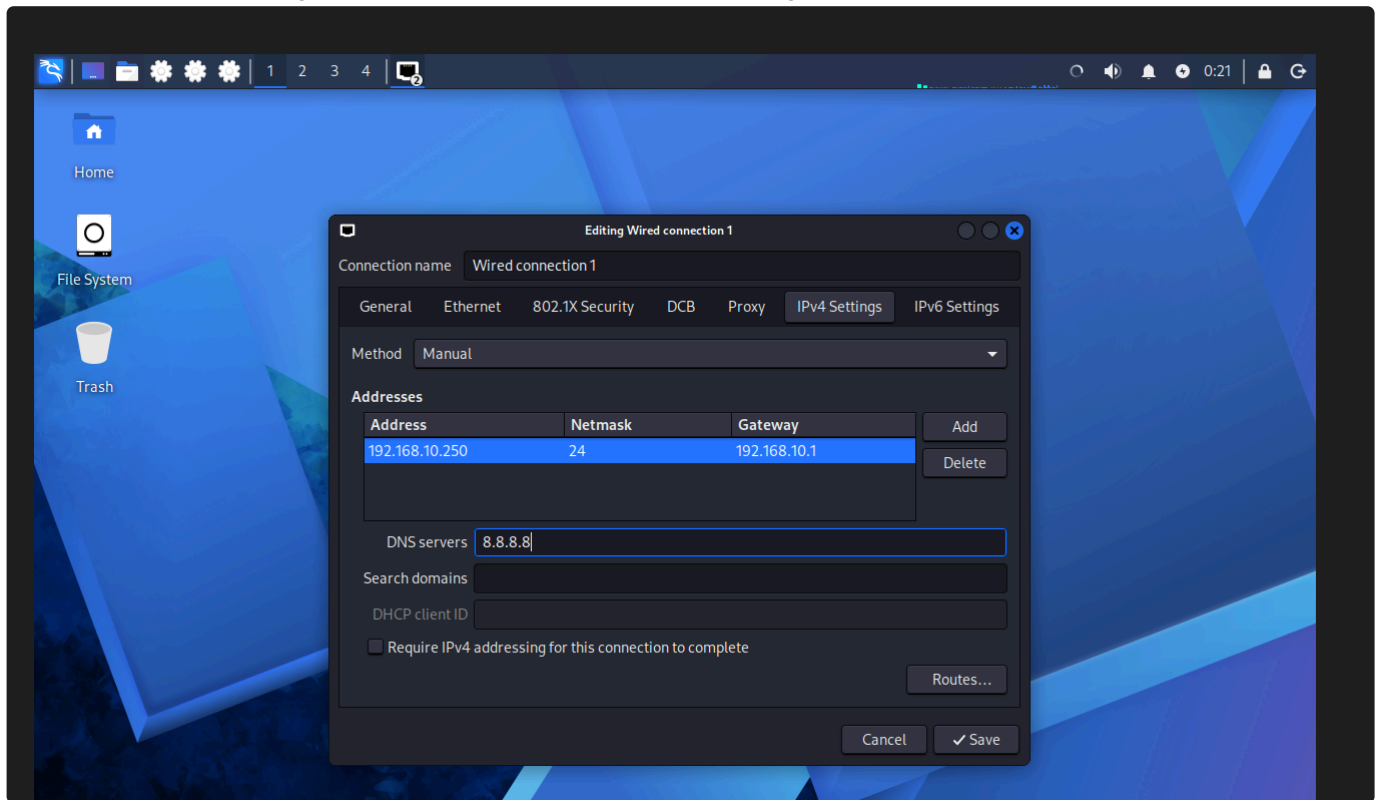
Once restarted select other user then log in with the credentials that we created in the domain



(optional but after the successful log in i took a snapshot of all the vms)

Kali Linux brute force attack | Viewing Telemetry via splunk | Setting up Atomic Red team and running test

Open kali VM and assign Kali a static IP that matches the diagram, then save.

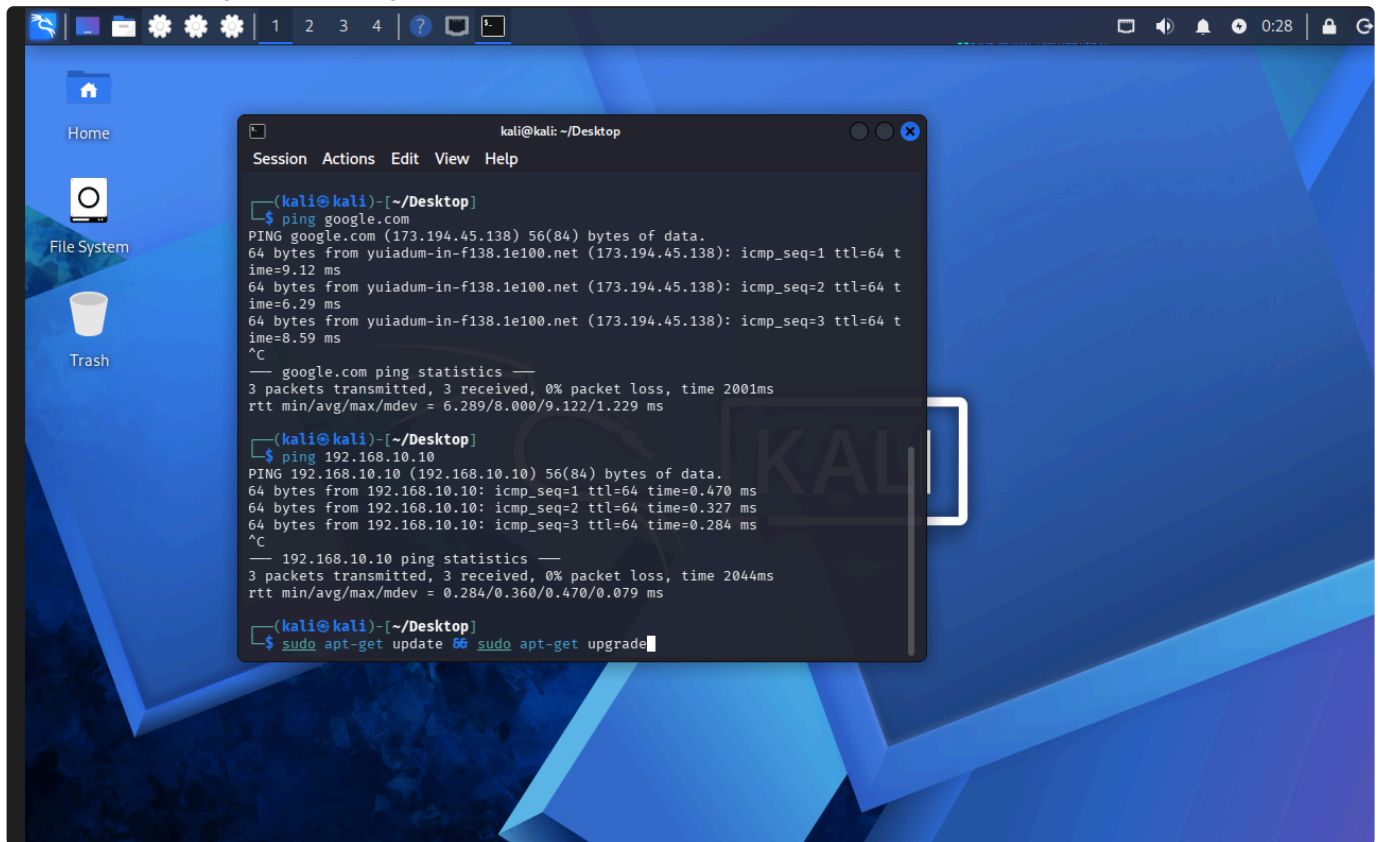


However even after you complete the step it wont actually show our new ip unless you disconnect the network interface and select wired connection 1

```
kali@kali: ~/Desktop
Session Actions Edit View Help
ault qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
inet6 fe80::b4a8:6068:adb1:4957/64 scope link noprefixroute
    valid_lft forever preferred_lft forever

(kali@kali)-[~/Desktop]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
ault qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
inet 192.168.10.250/24 brd 192.168.10.255 scope global noprefixroute eth0
    valid_lft forever preferred_lft forever
inet6 fe80::b4a8:6068:adb1:4957/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
```

run the usual ping test and upgrade/updates

A screenshot of a Kali Linux desktop environment. The desktop background is a blue and black geometric pattern. On the left sidebar, there are icons for 'Home', 'File System', and 'Trash'. A terminal window is open in the center, titled 'kali@kali: ~/Desktop'. The terminal shows the following commands and output:

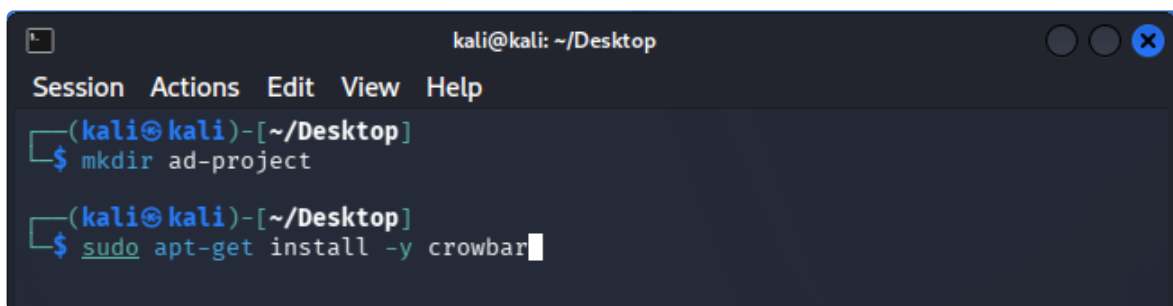
```
kali@kali: ~/Desktop
Session Actions Edit View Help

(kali@kali)-[~/Desktop]
$ ping google.com
PING google.com (173.194.45.138) 56(84) bytes of data.
64 bytes from yuiadum-in-f138.1e100.net (173.194.45.138): icmp_seq=1 ttl=64 t
ime=9.12 ms
64 bytes from yuiadum-in-f138.1e100.net (173.194.45.138): icmp_seq=2 ttl=64 t
ime=6.29 ms
64 bytes from yuiadum-in-f138.1e100.net (173.194.45.138): icmp_seq=3 ttl=64 t
ime=8.59 ms
^C
  google.com ping statistics
  3 packets transmitted, 3 received, 0% packet loss, time 2001ms
 rtt min/avg/max/mdev = 6.289/8.000/9.122/1.229 ms

(kali@kali)-[~/Desktop]
$ ping 192.168.10.10
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.
64 bytes from 192.168.10.10: icmp_seq=1 ttl=64 time=0.470 ms
64 bytes from 192.168.10.10: icmp_seq=2 ttl=64 time=0.327 ms
64 bytes from 192.168.10.10: icmp_seq=3 ttl=64 time=0.284 ms
^C
  192.168.10.10 ping statistics
  3 packets transmitted, 3 received, 0% packet loss, time 2044ms
 rtt min/avg/max/mdev = 0.284/0.360/0.470/0.079 ms

(kali@kali)-[~/Desktop]
$ sudo apt-get update && sudo apt-get upgrade
```

Create a new directory for our attack name ad-project using mkdir and then install crowbar which will be used for the brute force attack. Crowbar is a brute force tool which supports OPENVPN, RDP, SSH Private keys and VNC keys

A screenshot of a Kali Linux terminal window. The terminal is titled 'kali@kali: ~/Desktop'. It shows the following commands and output:

```
kali@kali: ~/Desktop
Session Actions Edit View Help

(kali@kali)-[~/Desktop]
$ mkdir ad-project

(kali@kali)-[~/Desktop]
$ sudo apt-get install -y crowbar
```

Change to wordlist directory to find rockyou and unzip the file using gunzip. Move the file to the new directory that we created. head back to the directory we created and run ls -lh to list files with detail in human readable format

```
kali@kali: /usr/share/wordlists
Session Actions Edit View Help
(kali@kali)-[/usr/share/wordlists]
$ ls
dirb          fasttrack.txt  legion        rockyou.txt.gz  wifite.txt
dirbuster     fern-wifi     metasploit    sqlmap.txt
dnsmap.txt    john.lst      nmap.lst     wfuzz

(kali@kali)-[/usr/share/wordlists]
$ sudo gunzip rockyou.txt.gz

kali@kali: ~/Desktop/ad-project
Session Actions Edit View Help
(kali@kali)-[~/Desktop/ad-project]
$ ls -lh
total 134M
-rw-r--r-- 1 kali kali 134M Sep 21 09:29 rockyou.txt

(kali@kali)-[~/Desktop/ad-project]
$
```

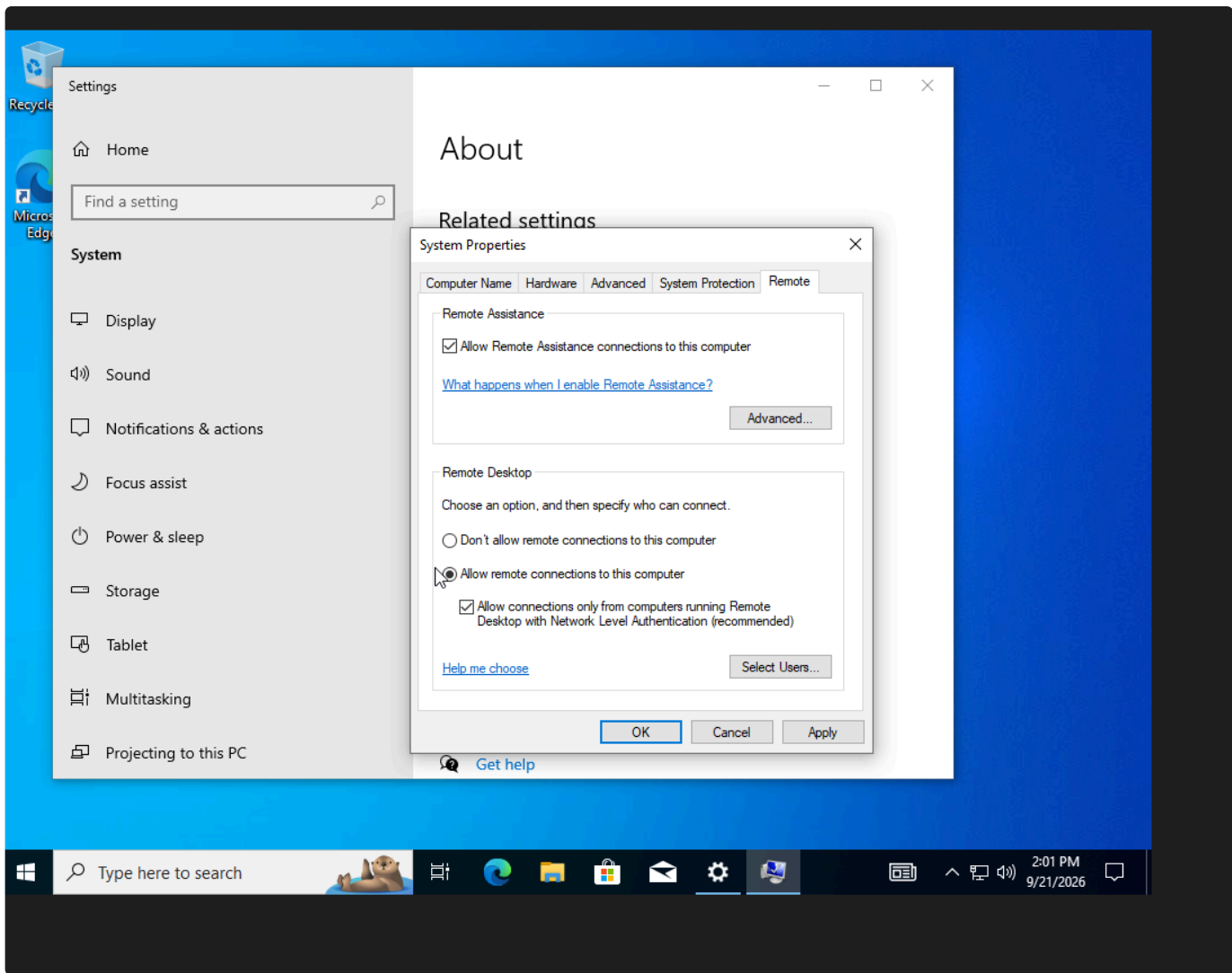
Only use 20 lines of the file since its 134mb and it will take a while. to do so use "head -n 20 rockyou.txt ".
move output in to password. txt. Add the secret password using nano

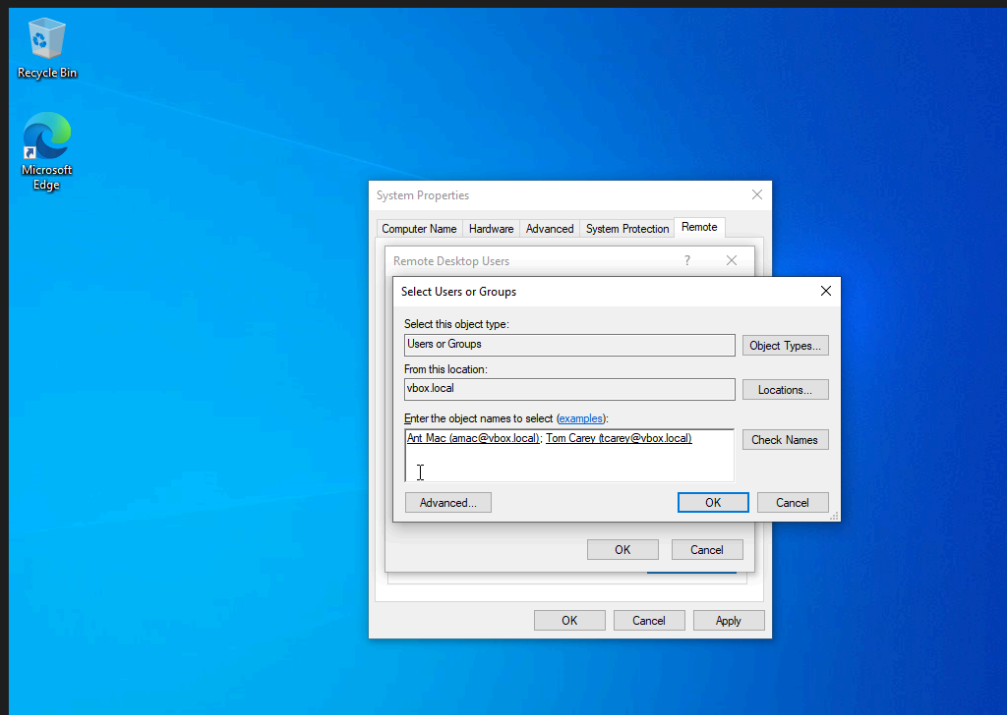
```
(kali@kali)-[~/Desktop/ad-project]
$ head -n 20 rockyou.txt
123456
12345
123456789
password
iloveyou
princess
1234567
rockyou
12345678
abc123
nicole
daniel
babygirl
monkey
lovely
jessica
654321
michael
ashley

(kali@kali)-[~/Desktop/ad-project]
$ head -n 20 rockyou.txt > passwords.txt
```

```
kali@kali: ~/Desktop/ad-project
Session Actions Edit View Help
GNU nano 9.2 passwords.txt *
123456
12345
123456789
password
iloveyou
princess
1234567
rockyou
12345678
abc123
nicole
daniel
babygirl
monkey
lovely
jessica
654321
michael
ashley
qwerty
Spring2027!
```

head back to target pc to enable rdp and add the users we created in active directory and apply





head back to kali and use crowbar -b rdp -u "user" -C passwords.txt -s 192.168.10.100/32
-u account of interest
-b to specify service
-C password list
-s source ip
we use /32 over 24 since we only want to target this ip

```
kali@kali: ~/Desktop/ad-project
Session Actions Edit View Help
(kali@kali)-[~/Desktop/ad-project]
$ crowbar -b rdp -u tcarey -C passwords.txt -s 192.168.10.100/32
```

Check splunk now and search index=endpoint tcarey

The screenshot shows the Splunk web interface. The search bar contains the query `search%20index%3Dendpoint%20tcarey...`. The results are displayed in a table with columns: Time, Event, and Log. The first result is for 9/21/26 at 8:04:39.650 PM. The event details include:

- LogName=Security
- EventCode=4625
- EventType=0
- ComputerName=target-pc.vbox.local
- SourceName=Microsoft Windows security auditing.
- Type=Information
- RecordNumber=1663
- Keywords=Audit Failure
- TaskCategory=Logon
- OpCode=Info
- Message=An account failed to log on.

The Subject section shows:

- Security ID: S-1-0-0
- Account Name: -
- Account Domain: -
- Logon ID: 0x0
- Logon Type: 3

when you search up the eventcode4625 you will see that its the log event id that states An account failed to log on

Setting up Atomic Red team

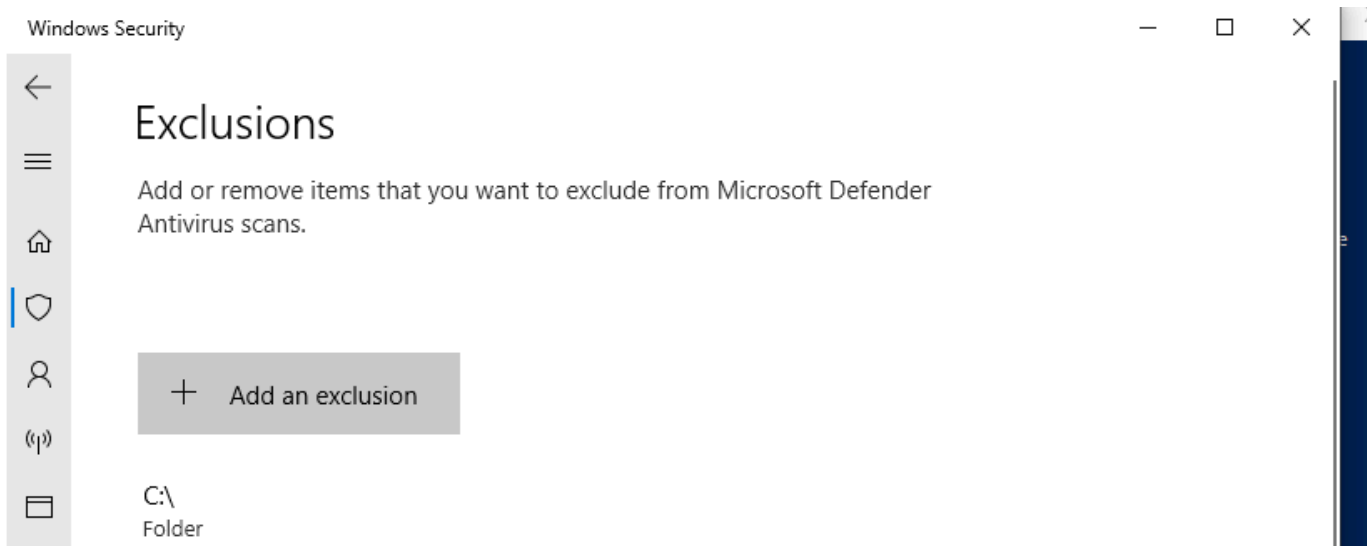
run powershell as admin and run "Set-ExecutionPolicy Bypass CurrentUser". In Windows security add exclusion to your c drive. Run "IEX (IWR '<https://raw.githubusercontent.com/redcanaryco/invoke-atomicredteam/master/install-atomicredteam.ps1>' -UseBasicParsing); Install-AtomicRedTeam -getAtomsics" on powershell.

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

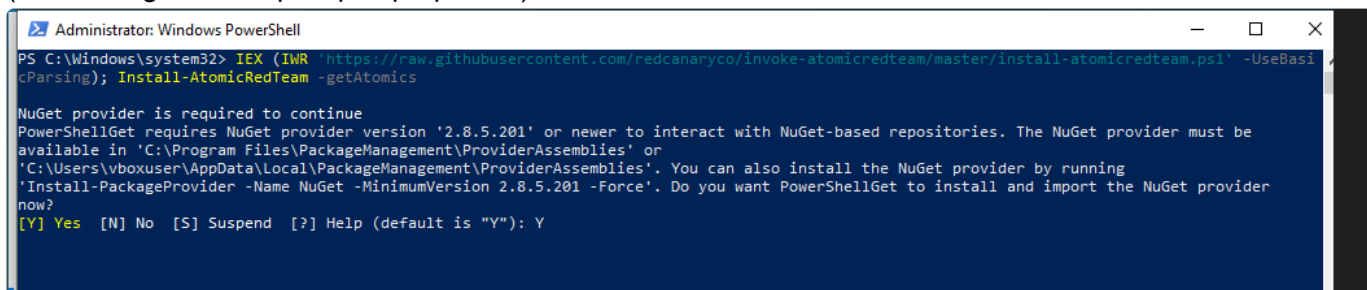
Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> Set-ExecutionPolicy Bypass CurrentUser

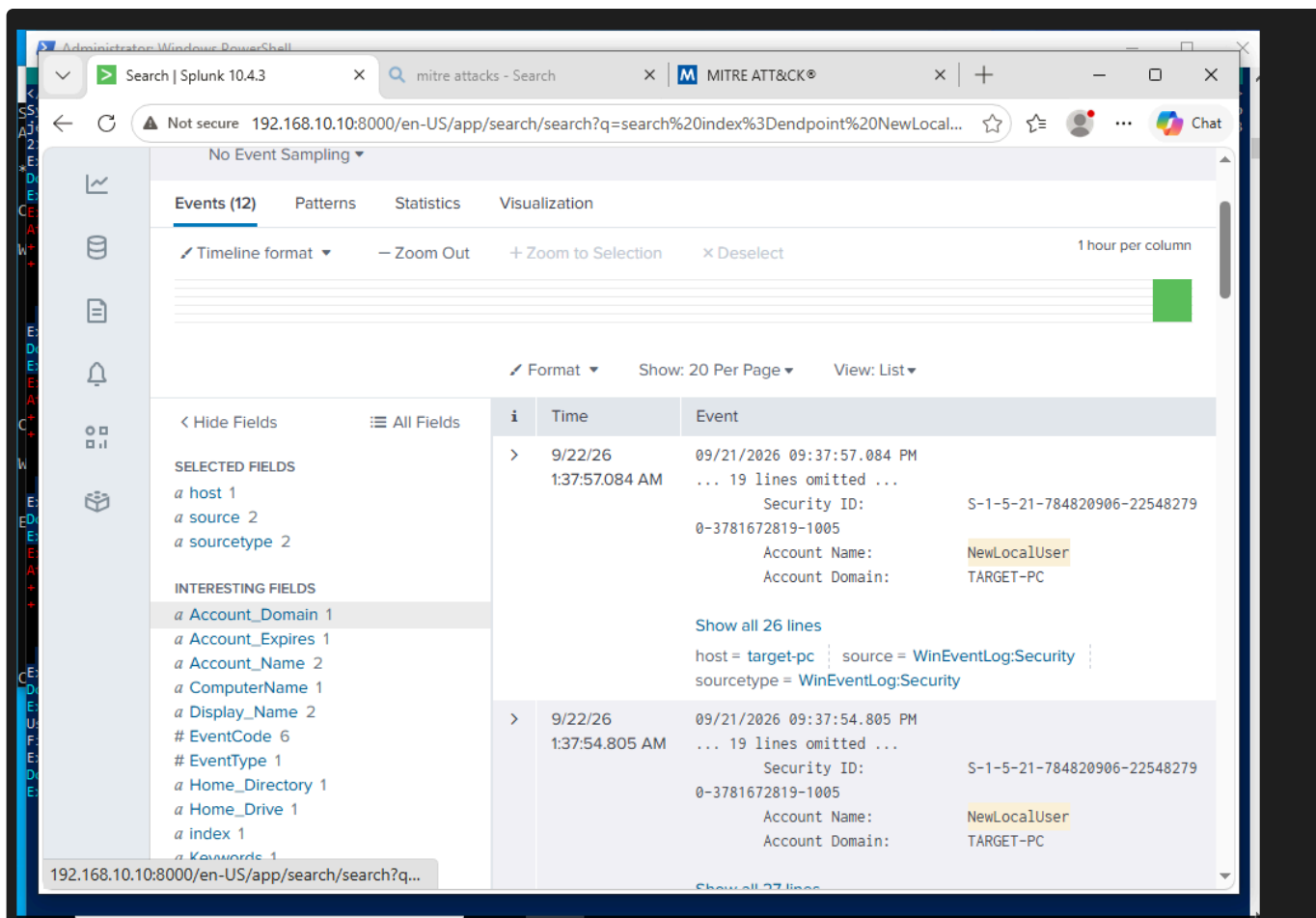
Execution Policy Change
The execution policy helps protect you from scripts that you do not trust. Changing the execution policy might expose
you to the security risks described in the about_Execution_Policies help topic at
https://go.microsoft.com/fwlink/?LinkID=135170. Do you want to change the execution policy?
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "N"): Y
PS C:\Windows\system32>
```



(note changed to dhcp in ipv4 properties)



Inside the new atomic folder we see technique ids that frame back to MITRE attack framework. for the example I'll use create local account "T1136.001". Run "Invoke-AtomicTest T1136.001" which will create telemetry based on running an local account. Telemetry is the automatic collection and transmission of data from a system, so it can be monitored, analyzed, or reported on remotely, without someone manually gathering it.



Conclusion

This lab gave me hands on experience building a small enterprise style network from the ground up. I set up a Windows Server 2022 domain controller, promoted it to Active Directory, and joined a Windows 10 client to the domain. I also built a Splunk server to collect and analyze logs, using Sysmon and the Universal Forwarder to pull detailed endpoint activity from the client into Splunk for searching and detection. On the offensive side, I used a Kali Linux machine to simulate an attacker, running brute force attempts against RDP and setting up Atomic Red Team to generate test activity that Splunk could then detect and log.

Along the way I ran into a lot of real problems that didn't come from a guide, they came from actually building the environment myself. I dealt with a corrupted Windows Server ISO, a Splunk login that broke because of a password length requirement, a Universal Forwarder service that got stuck mid restart, DNS resolution failing after joining the domain, and RDP connections failing due to NLA and Kerberos issues. Each of these forced me to slow down, read error messages carefully, and work through the actual root cause instead of guessing.

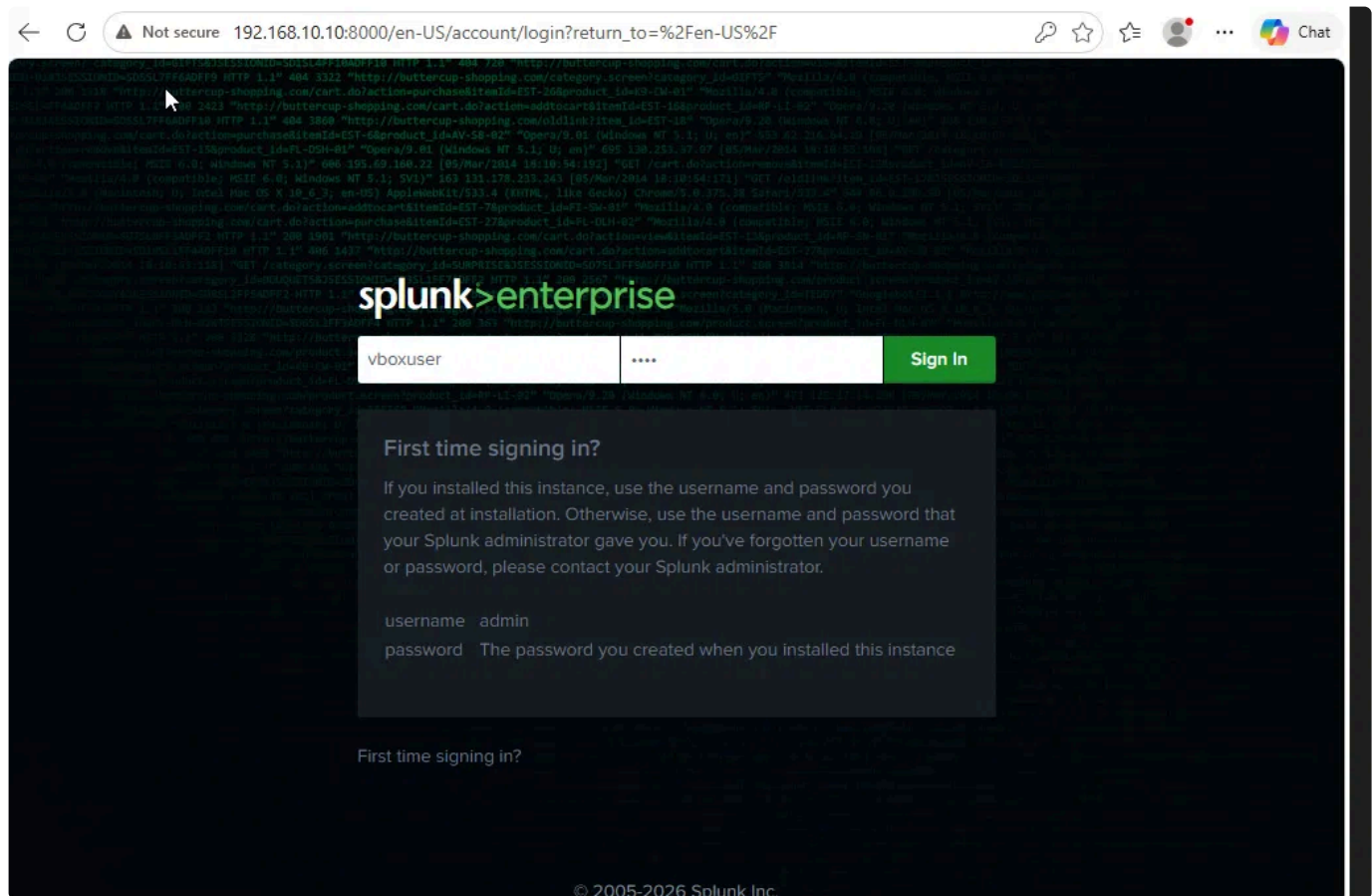
What this lab really taught me is that setting up the tools is only half the work. The other half is understanding how everything connects, how DNS ties into domain authentication, how Sysmon and the Universal Forwarder work together to get useful data into Splunk, and how a small misconfiguration in one place can cause a completely different looking error somewhere else. I also learned the value of documenting problems as they happen, since writing out each issue and how I solved it forced me to actually understand the fix instead of just moving past it.

Overall, this project pushed me to build, break, and fix a realistic small network on my own, which is exactly the kind of hands on experience I wanted. It gave me a much better understanding of how Active Directory, endpoint logging, and basic attack simulation fit together in a real environment.

Issues that occurred during the lab and how i troubleshooted them

SPLUNK LOGIN ISSUE

Error occurred when i initially tried to log into splunk



1. Attempted to log in using vboxuser and the Ubuntu OS credentials, assuming Splunk shared the same login as the Linux server
2. Checked /opt/splunk/etc/passwd to confirm which admin account existed on the system
3. Confirmed the file showed a vboxuser account, but login still failed using that username and the password I believed was correct
4. Attempted a password reset using splunk edit user with the default -auth admin:changeme flag
5. Decided to fully reset the admin account by renaming/removing the passwd file so Splunk would be forced to create a new one
6. Restarted Splunk and expected either a "create new user" setup page or the classic admin\changeme default login, neither of which worked
7. Verified Splunk was actually running with splunk status (confirmed it was, ruling out a service issue)
8. Ruled out browser caching by hard refreshing and testing in an InPrivate/incognito window, same result
9. Researched and learned that Splunk versions 8.1.1+ removed the old admin:changeme default entirely for security reasons, and instead require an admin account to be created manually before first startup

10. Created a user-seed.conf file under /opt/splunk/etc/system/local/ to define a new admin username and password ahead of Splunk's next startup
11. Restarted Splunk again, but the passwd file still wasn't generated
12. Checked file ownership and permissions on user-seed.conf to rule out a root vs vboxuser permission mismatch
13. Reviewed the exact contents of the seed file and discovered the original password was too short to meet Splunk's minimum password length requirement, causing Splunk to silently reject it
14. Updated the password in user-seed.conf to meet the minimum character requirement
15. Restarted Splunk one final time, which successfully generated the passwd file and allowed login with the new admin credentials

Root cause The seed file's password did not meet Splunk's minimum password complexity/length requirement, so Splunk silently ignored the file instead of creating the admin account, which made the issue look like a permissions or caching problem for several steps before the actual cause was identified.

(NOT the new/real password but similar in to the old login length)



```
GNU nano 8.7.1 /opt/splunk/etc/system/local/user-seed.conf *
[user@rtr6]
USERNAME = admin
PASSWORD = pass
```

The screenshot shows a terminal window with the nano text editor open. The title bar indicates the file path is /opt/splunk/etc/system/local/user-seed.conf *. The editor content shows the configuration for the admin user: USERNAME = admin and PASSWORD = pass. The bottom status bar displays various nano editor shortcuts such as Help, Exit, Write Out, Read File, Where Is, Replace, Cut, Paste, Execute, Justify, Location, Go To Line, Undo, Redo, Set Mark, Copy, To Bracket, Where Was, Previous, and Next.

```

vboxuser@Splunk0:~$ sudo /opt/splunk/bin/splunk start
Splunk> Finding your faults, just like mom.
Checking prerequisites...
  Checking http port [8000]: open
  Checking mgmt port [8089]: open
  Checking appserver port [127.0.0.1:8065]: open
  Checking kvstore port [8191]: open
  Checking configuration... Done.
New certs have been generated in '/opt/splunk/etc/auth'.
  Checking critical directories... Done
  Checking indexes...
    Validated: _audit _configtracker _dm_summary _dsappevent _dsclient _dsphonehome _internal _introspection _metrics _metrics_ro
efishbucket history main summary
    Done
  Checking filesystem compatibility... Done
  Checking conf files for problems...
    Done
  Checking default conf files for edits...
  Validating installed files against hashes from '/opt/splunk/splunk-10.4.3-4174a2deda5d-linux-amd64-manifest'
  All installed files intact.
    Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Done

Waiting for web server at http://127.0.0.1:8000 to be available.....

```

Issue: No logs appearing in Splunk under index=endpoint despite configuring inputs.conf

Steps taken to diagnose and resolve:

1. Ran a search for index=endpoint in Splunk and got 0 events, confirming no data was being received
2. Verified the endpoint index actually existed and was Active under Settings > Indexes, ruling out a missing index as the cause
3. Reviewed the inputs.conf file on the Windows 11 client to confirm it was correctly set up to monitor Application, Security, System, and Sysmon logs and route them to the endpoint index
4. Checked outputs.conf on the Windows client to confirm the forwarder was pointed at the Splunk server's IP and port 9997
5. Found an invalid extra stanza at the bottom of outputs.conf ([tcpout-server://...]), which is not valid Splunk syntax, and removed it
6. Attempted to restart the SplunkForwarder service using `net stop SplunkForwarder`, which failed with "The service is starting or stopping. Please try again later."
7. Tried restarting directly from the Windows Services panel, but the right-click menu was greyed out, meaning the service was unresponsive
8. Used an elevated command prompt to force the service to stop with `sc stop SplunkForwarder`, which returned an error ("The pipe has been ended"), indicating the service was in a broken state
9. Ran `sc start SplunkForwarder` to force a fresh start attempt, which returned a `START_PENDING` state with an active PID, confirming the process was launching
10. Waited briefly, then confirmed the service fully started using `sc query SplunkForwarder`, which returned `STATE: 4 RUNNING`

11. Checked Splunk's Forwarder Management page to confirm the Windows client was now checking in successfully
12. Re-ran the index=endpoint search with an updated time range and confirmed events were now appearing

Root cause: A malformed stanza in outputs.conf combined with the SplunkForwarder service getting stuck in a broken start/stop state prevented the forwarder from establishing a connection to the Splunk server, so no log data was ever transmitted.

